



Master's Thesis

A Systematic Comparison of Classical Statistical Pre-processing and AI-based Representation Learning for Business Intelligence

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Declaration of authorship

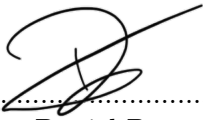
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I acknowledge the use of Claude (Anthropic) throughout this work, in the following roles: implementation of the experimental code, literature search and retrieval, translation between German and English, and the drafting, revision and proofreading of the running text. The research question, the experimental design, the interpretation of the results and every substantive claim made in this thesis are my own, as is the responsibility for them. All reported figures originate from experiments I conducted and are reproducible from the published repository.

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Abstract

Pre-processing consumes a large share of the effort in predictive Business Intelligence, yet it is rarely studied as a factor in its own right. Deep representation learning promises to replace manual feature engineering with learned features, but the comparative literature evaluates such architectures as classifiers, which confounds the quality of the representation with that of the classification head. This work isolates the pre-processing effect. Six pipelines, one of them instantiated in two variants, are each used strictly as pre-processing operators feeding five downstream classifiers. Three classical: A baseline of imputation, encoding and scaling, a statistical extension with winsorization, Box-Cox transformation and finally principal component analysis and domain feature engineering. As well as three AI-based pipelines: denoising autoencoder, variational autoencoder and FT-Transformer. The evaluation covers the UCI Bank Marketing and Online Retail datasets, uses PR-AUC as the primary metric given class imbalance, and applies 5×3 repeated stratified cross-validation across 70 pipeline-model-dataset combinations, with differences judged by their magnitude against the fold standard deviation rather than by significance testing.

No AI-based pipeline surpassed even the plain classical baseline under the best-of-five reading. The strongest trailing the leading classical pipeline by 0.016 on both datasets. For two individual classifiers on Bank Marketing it does edge past that baseline. The feature-engineering pipeline achieved the highest PR-AUC on both datasets (0.4728 and 0.7797), but the two datasets support that ranking to a different degree: on Bank Marketing its lead over the baseline is 0.0025 and holds for only two of the five classifiers, so the comparison is left undecided there, whereas on Online Retail it leads by 0.0090 in the model mean and for all five. An ablation decomposing each pipeline into its constituent operations locates that lead entirely in the polynomial interaction terms and not in the domain-specific features, which contribute nothing on either dataset, so the operation that pays off is generic and requires no subject-matter knowledge. The same ablation showed that feature scaling alone accounted for three quarters of what the complete baseline achieved and that several complete pipelines performed worse than their best individual step. Under that protocol, which grants no pipeline a tuning advantage, the supervised encoder of Pipeline F was the strongest single operation of the study, inverting the ordering obtained with tuned downstream models and showing that the deficit of the learned representations depends on how the comparison is set up. Robustness resisted the ordering: on Bank Marketing the feature-engineering pipeline proved the least robust configuration, degrading by 12.0 % on average under injected missingness, outliers and label noise, because its derived features propagate corruption from their inputs. Within the family of learned representations the training objective governed robustness far more than the architecture, the supervised FT-Transformer losing 6.6 % under missingness not at random against 27.7 % and 34.0 % for the two autoencoders.

For Business Intelligence practice the results support a clear order of priority. Imputation, encoding and above all scaling come first, since they account for most of the achievable benefit. Polynomial interaction terms on the standardized numeric features are the one elaboration that repaid its effort, provided input quality is monitored, since the resulting pipeline delivers the best accuracy but is also the most corruption-sensitive. Learned representations are not recommended for problems of this size and structure, because they offer no accuracy advantage under a tuned classifier and remove the interpretability on which business decisions depend.

Kurzfassung

Preprocessing bindet einen erheblichen Teil des Aufwands in der prädiktiven Business Intelligence, wird jedoch selten als eigenständiger Einflussfaktor untersucht. Representation Learning verspricht, manuelles Feature Engineering zu ersetzen, doch die Vergleichsliteratur bewertet solche Architekturen als Klassifikatoren und vermengt damit die Qualität der Repräsentation mit der des Klassifikationskopfes. Diese Arbeit isoliert den Preprocessing-Effekt. Sechs Pipelines, eine davon in zwei Varianten, dienen ausschließlich als Preprocessing-Operatoren für fünf nachgelagerte Klassifikatoren. Drei Klassische: Basis aus Imputation, Encoding und Skalierung, eine statistische Erweiterung mit Winsorizing, Box-Cox Transformation und Hauptkomponentenanalyse, sowie domänenspezifisches Feature Engineering. Außerdem drei KI-basierte: Denoising Autoencoder, Variational Autoencoder und FT-Transformer. Die Evaluation umfasst die UCI-Datensätze Bank Marketing und Online Retail, verwendet PR-AUC als Primärmetrik angesichts der Klassenimbalance und wendet eine 5×3 wiederholte stratifizierte Kreuzvalidierung über 70 Kombinationen an. Unterschiede werden an ihrer Größe gemessen an der Fold-Standardabweichung beurteilt, nicht über Signifikanztests.

Keine KI-basierte Pipeline übertraf nach der Best-of-Five-Regel die klassische Basislinie. Die stärkste blieb auf beiden Datensätzen um 0,016 hinter der führenden klassischen Pipeline zurück, bei zwei einzelnen Klassifikatoren auf Bank Marketing liegt sie knapp davor. Die Feature-Engineering-Pipeline erzielte auf beiden Datensätzen die höchste PR-AUC (0,4728 bzw. 0,7797), doch die Datensätze stützen diese Rangfolge unterschiedlich stark: Auf Bank Marketing beträgt der Vorsprung gegenüber der Basislinie 0,0025 und besteht bei zwei der fünf Klassifikatoren, der Vergleich bleibt dort also unentschieden. Auf Online Retail führt sie im Modellmittel um 0,0090 und bei allen fünf. Eine Ablationsstudie verortet diesen Vorsprung vollständig in den polynomiellen Interaktionstermen und nicht in den domänenspezifischen Merkmalen, die auf beiden Datensätzen nichts beitragen. Die wirksame Operation ist damit generisch und setzt kein Fachwissen voraus. Dieselbe Ablation zeigte, dass allein die Skalierung drei Viertel dessen erbrachte, was die vollständige Basislinie erreichte, und dass mehrere vollständige Pipelines schlechter abschnitten als ihr bester Einzelschritt. Unter diesem Protokoll, das keiner Pipeline einen Tuningvorteil einräumt, war der überwacht trainierte Encoder der Pipeline F die stärkste Einzeloperation der Studie. Das kehrt die mit getunten Modellen erzielte Rangfolge um: Der Rückstand hängt von der Anlage des Vergleichs ab. Bei der Robustheit kehrte sich das Bild: Auf Bank Marketing war die Feature-Engineering-Pipeline unter injizierten Fehlwerten, Ausreißern und Label-Noise mit 12,0 % Verlust die am wenigsten robuste Konfiguration, da ihre abgeleiteten Merkmale Korruption aus den Basismerkmalen weitergeben. Innerhalb der gelernten Repräsentationen bestimmte das Trainingsziel die Robustheit stärker als die Architektur: Der FT-Transformer verlor unter MNAR 6,6 % gegenüber 27,7 % und 34,0 % der Autoencoder.

Für die Praxis der Business Intelligence ergibt sich eine klare Reihenfolge. An erster Stelle stehen Imputation, Encoding und vor allem Skalierung, die den Großteil des erreichbaren Nutzens ausmachen. Als einzige weiterführende Maßnahme lohnten polynomielle Interaktionsterme auf den standardisierten numerischen Merkmalen den Zusatzaufwand, sofern die Datenqualität überwacht wird, denn die entsprechende Pipeline liefert die beste Genauigkeit, ist zugleich aber am korruptionsanfälligsten. Gelernte Repräsentationen sind für Probleme dieser Größe und Struktur nicht zu empfehlen, da sie unter einem getunten Klassifikator keinen Genauigkeitsvorteil bieten und die Interpretierbarkeit beseitigen, auf die betriebliche Entscheidungen angewiesen sind.

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Key Terms

Business Intelligence

Data pre-processing

Representation learning

Tabular data

Imbalanced classification

Feature engineering

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Chapter 1

Introduction

Predictive models in Business Intelligence rest on data that has been prepared for them. This preparation is rarely the object of study, yet the evidence suggests it matters more than the choice of learning algorithm. This work examines whether learned representations can replace the manual preparation that currently dominates practice, and it does so by isolating the preparation step from everything that surrounds it.

1.1 Motivation and Problem Statement

Business Intelligence has shifted over the past two decades from retrospective reporting toward predictive decision support [1, 2]. Where dashboards once summarized what had happened, contemporary BI systems are expected to estimate what will happen: which customers will churn, which prospects will convert, which accounts warrant intervention. The underlying data is almost invariably tabular (transaction records, customer master data, campaign logs), and the analytical questions are almost invariably imbalanced, because the events worth predicting are by their nature rare [3].

Within the modeling workflow, pre-processing occupies an unusual position. It consumes a large share of the practical effort, it is rarely the subject of systematic study, and it is frequently treated as a solved problem to be dispatched with a standard recipe of imputation, encoding and scaling. The first of these is documented. In the interview study of Kandel et al. [4] across 35 analysts in 25 organisations, data wrangling emerges as the most tedious and time-consuming phase of the analysis process, with one respondent reporting more than half of their working time spent integrating, cleansing and transforming data before any analysis begins. Sambasivan et al. [5] arrive at the complementary finding that this effort is systematically undervalued relative to model work, which is a plausible reason why an activity of that magnitude has attracted so little comparative study. Yet the evidence suggests that this step is more decisive than the choice of algorithm. McElfresh et al. [6] compare 19 algorithms across 176 datasets and find that dataset characteristics frequently dominate the choice of learner, and the work of Grinsztajn et al. [7] and Shwartz-Ziv and Armon [8] shows that carefully applied gradient boosting on well-prepared tabular features remains competitive with, and often superior to, deep architectures.

At the same time, representation learning offers a different proposition: rather than construct-

ing features by hand, learn them from the data [9]. Denoising autoencoders [10], variational autoencoders [11] and, more recently, transformer architectures adapted to tabular input [12, 13] promise to discover structure that manual feature engineering would miss. For a Business Intelligence practitioner this is an attractive proposition, because domain feature engineering is expensive, requires scarce subject-matter knowledge, and does not transfer between problems.

The two positions are in tension, and the tension is not merely about accuracy. Classical pipelines are transparent: a marketing manager can be told that employment level drives subscription probability and can act on that. Learned representations are opaque by construction, since a latent dimension carries no business meaning. Classical pipelines are cheap, while training a transformer encoder is not. In a field where models inform human decisions and increasingly require justification, an accuracy comparison alone is insufficient.

What is missing is a study that isolates the pre-processing effect itself. The existing comparative literature evaluates deep architectures as *classifiers*, which confounds the quality of the learned representation with the quality of the classification head. If representation learning is to be assessed as a pre-processing strategy, which is how a BI team would actually deploy it by feeding learned features into an established downstream model, it must be compared against classical pre-processing under an otherwise identical protocol. Neither is the effect typically decomposed: aggregate performance figures do not reveal whether a pipeline’s advantage comes from its sophisticated components or from an elementary step that any implementation would include.

1.2 Research Questions

This work addresses two research questions. They correspond to the two considerations that determine whether a pre-processing strategy is adopted in practice, namely whether it predicts better and whether it survives imperfect data.

RQ1: Predictive performance How do classical pre-processing pipelines and AI-based representation-learning pipelines differ in predictive performance on real business datasets?

RQ2: Robustness Which pre-processing strategy exhibits the greatest robustness to missing values, outliers and label noise?

1.3 Approach and Contribution

Six pre-processing pipelines are compared. Three are classical: a baseline of imputation, encoding and scaling, a statistically extended variant adding winsorization, Box-Cox transformation and principal component analysis, and a domain feature engineering variant. Three are AI-based: a denoising autoencoder, a variational autoencoder and an FT-Transformer. Each is used strictly as a pre-processing operator whose output is passed to five downstream classifiers, so that the representation is the only quantity that varies. The evaluation spans two public business datasets, uses PR-AUC as the primary metric in view of class imbalance [14, 15], and applies 5×3 repeated stratified cross-validation. Differences are judged by their magnitude in PR-AUC read against the fold standard deviation. A Wilcoxon signed-rank test with Bonferroni correction [16] was computed as well, but its assumptions are not met by the

fold structure, so its results are reported for completeness only and carry no conclusion, which Section 4.3 sets out.

This work makes three contributions:

First, it provides a systematic comparison of classical and learned pre-processing under an identical downstream protocol, isolating the pre-processing effect from the model effect.

Second, it decomposes each pipeline into its constituent operations through an ablation study, which identifies where the performance actually originates from.

Third, it extends the comparison beyond accuracy to the dimensions that govern adoption in practice: robustness to data quality problems and the interpretability of the resulting features.

1.4 Scope and Delimitation

The study examines pre-processing as an isolated factor. Complete AutoML systems are excluded because they couple pre-processing and model selection, which is the confounding this work is designed to avoid. Architectures such as TabNet [17] that operate as end-to-end classifiers are not considered. The FT-Transformer is employed as a feature extractor, not as a complete model. Generative approaches to data synthesis and non-tabular modalities are outside the scope.

The evaluation covers two datasets. This permits an assessment of whether findings hold across structurally different problems, but does not establish generality across the full range of business data.

1.5 Structure of This Work

Chapter 2 develops the theoretical background: the Business Intelligence context, the classical pre-processing operations, the representation-learning architectures, and the evaluation methodology. Chapter 3 describes the experimental design, the datasets, the six pipelines and the evaluation protocol, including the leakage prevention measures that proved essential to obtaining valid results. Chapter 4 presents and discusses the results along the two research questions. Chapter 5 summarizes the findings, states their limitations and outlines directions for further work.

Chapter 2

Theoretical Background

This chapter establishes the conceptual foundations on which the comparison in this work rests. Section 2.1 situates predictive modeling within Business Intelligence and derives the requirements that follow from that context. Section 2.2 treats class imbalance, which determines the choice of evaluation metric. Section 2.3 develops the classical pre-processing operations, Section 2.4 the representation-learning architectures, and Section 2.5 the downstream classifiers. Section 2.6 covers the evaluation methodology, Section 2.7 the taxonomy of information leakage, and Section 2.8 positions the study within the existing literature.

2.1 Business Intelligence and Predictive Analytics

Business Intelligence supplies the context in which the pipelines of this work are evaluated. The requirements that follow from that context, in particular the weight given to interpretability, robustness and cost, differ from those of a purely predictive research setting and shape the evaluation design.

2.1.1 Definition and Evolution

Business Intelligence denotes the set of technologies, processes and applications through which organizations collect, integrate and analyze data in order to support decision-making. Negash [1] characterizes BI as a system that combines data gathering, data storage and knowledge management with analytical tools in order to convert raw data into information relevant for decisions. Watson and Wixom [2] emphasize the organizational dimension: BI is not merely a technology stack but a capability, and its value depends on whether the resulting information actually alters decisions.

The classical architecture rests on a data warehouse that consolidates transactional source systems into a dimensionally modeled, query-optimized store [18]. On top of this foundation, three analytical layers are conventionally distinguished:

- **Descriptive analytics** answers what has happened. Reports and dashboards belong to this layer, and they continue to account for the majority of BI usage in practice.
- **Predictive analytics** answers what is likely to happen. This is where supervised

machine learning enters: churn prediction, response modeling, credit scoring, demand forecasting.

- **Prescriptive analytics** answers what should be done, combining predictions with optimization or simulation to recommend actions.

This work operates in the predictive layer. The distinction matters because the requirements differ from those of a purely predictive research setting. A BI model is embedded in an organizational process, so its outputs are consumed by people who must act on them, frequently justify those actions, and who will discard a model they do not trust. Three consequences follow, and all three recur in the evaluation design of this study.

Interpretability is a functional requirement, not a preference. A marketing department that receives a ranked contact list will ask why a particular customer ranks highly. If the answer cannot be given in terms of observable business quantities, the model can still be used but cannot inform strategy, and in regulated domains it may not be deployable at all. Rudin [19] argues that in high-stakes settings a post-hoc explanation of an opaque model is no substitute for a model that is interpretable to begin with, since the explanation is itself an approximation that may not reflect what the model does. This is the reason why the interpretability of the resulting feature space is treated as a result dimension in Chapter 4 rather than as an afterthought.

Robustness to data quality problems is a routine concern. BI systems draw on operational source systems that were designed for transaction processing, not for analysis. Missing values, coding conventions, sentinel values, duplicate records and silent schema changes are the normal case rather than the exception. Sambasivan et al. [5] document how such upstream defects propagate into compounding downstream failures that are detected late and are expensive to repair, a pattern they term data cascades. A pipeline whose performance depends on pristine input is therefore of limited practical value, which is why robustness to injected corruption forms the second research question of this work.

Maintenance burden constrains adoption. An approach that requires specialized expertise or a maintenance burden disproportionate to its benefit will not be adopted, however favorable its benchmark results. Sculley et al. [20] observe that the learning code is a small part of a deployed system and that most of the ongoing effort is caused by the surrounding data plumbing, of which pre-processing is a central component. A pre-processing strategy is therefore adopted or rejected on grounds that a benchmark table does not capture.

2.1.2 The Role of Pre-processing

Within the modeling workflow, pre-processing transforms raw source data into the numerical representation a learning algorithm requires. The operations involved are individually elementary (filling gaps, encoding categories, adjusting scales), but their composition determines what information the model can access at all.

Pre-processing occupies a curious position in the literature. It consumes a substantial share of practical project effort, yet it is rarely the object of systematic study. Comparative work concentrates on model architectures, with pre-processing held fixed and unreported. This is problematic for two reasons. First, if pre-processing choices influence outcomes as strongly as model choices, then model comparisons that fix one arbitrary pre-processing configuration are

not measuring what they claim to measure. Second, a practitioner deciding where to invest effort receives no guidance from a literature that treats the step as solved.

The evidence available suggests the influence is considerable. McElfresh et al. [6] compare 19 algorithms across 176 datasets and find that dataset characteristics frequently dominate the choice of algorithm, and the comparative studies of tabular deep learning discussed in Section 2.8 report that well-prepared features fed to gradient boosting remain a strong baseline against considerably more elaborate architectures.

2.2 Imbalanced Classification

In most BI prediction problems the event of interest is rare. Term deposit subscriptions, customer churn, fraud and equipment failure occur in a small fraction of cases. In the Bank Marketing dataset used here the positive rate is 11.3%. He and Garcia [3] survey the difficulties that follow. Two of them are relevant here.

First, accuracy becomes uninformative. A classifier that predicts the majority class unconditionally achieves 88.7% accuracy on Bank Marketing while being useless for its intended purpose. Any evaluation of an imbalanced problem must therefore use metrics that are sensitive to minority-class performance. Second, the decision threshold loses its natural interpretation. At the conventional threshold of 0.5, a model trained on imbalanced data will rarely predict the positive class, even when its ranking of instances is informative. Because BI applications typically consume a *ranking* (the top decile of a contact list, the highest-risk accounts), threshold-independent metrics are both more appropriate and more closely aligned with the intended use.

Both difficulties are addressed here through the choice of metric rather than by altering the training data. Resampling methods such as SMOTE [21] are themselves pre-processing operations with a substantial effect on downstream performance, so applying one inside some pipelines and not others would confound exactly the comparison this study is designed to make.

2.3 Classical Statistical Pre-processing

The operations described in this section make up Pipelines A to C. They are presented individually because the ablation study of Section 4.4 measures each of them in isolation, and the theory developed here predicts several of the results reported there.

2.3.1 Missing Value Imputation

Little and Rubin [22] establish the taxonomy that governs how missing data may be treated. It distinguishes three mechanisms by what the missingness depends on.

Missing completely at random (MCAR). The probability that a value is missing is independent of both the observed and the unobserved data. A sensor that fails randomly produces MCAR gaps. Under MCAR the observed cases are a random subsample of the full data, so simple imputation by a measure of central tendency introduces no systematic bias, although it understates variance.

Missing at random (MAR). The probability of missingness depends on observed variables but not on the missing value itself. If high-income respondents are less likely to report their age, and income is observed, the age missingness is MAR. Conditional imputation using the observed covariates can correct the bias in principle. Multiple imputation [23] is the standard approach.

Missing not at random (MNAR). The probability of missingness depends on the unobserved value itself. If high-income respondents specifically decline to report their income, the missingness is MNAR. No imputation based solely on the observed data can correct this, because the distribution of the missing values differs systematically from that of the observed ones. Median imputation is then biased by construction, since the imputed value is drawn from a distribution that excludes the very region where the true values lie.

Fig. 2.1 makes the consequence visible on a simulated feature from which 30 % of the values have been removed under each of the three mechanisms. Under MCAR, Fig. 2.1(a), the median of the values that remain is indistinguishable from the median of the complete data. Under MAR, Fig. 2.1(b), the two separate, and under MNAR, Fig. 2.1(c), the separation is largest. Only in the last case is any imputation drawn from the observed values biased by construction, which is why the robustness experiments of this work treat MNAR as the decisive mechanism.

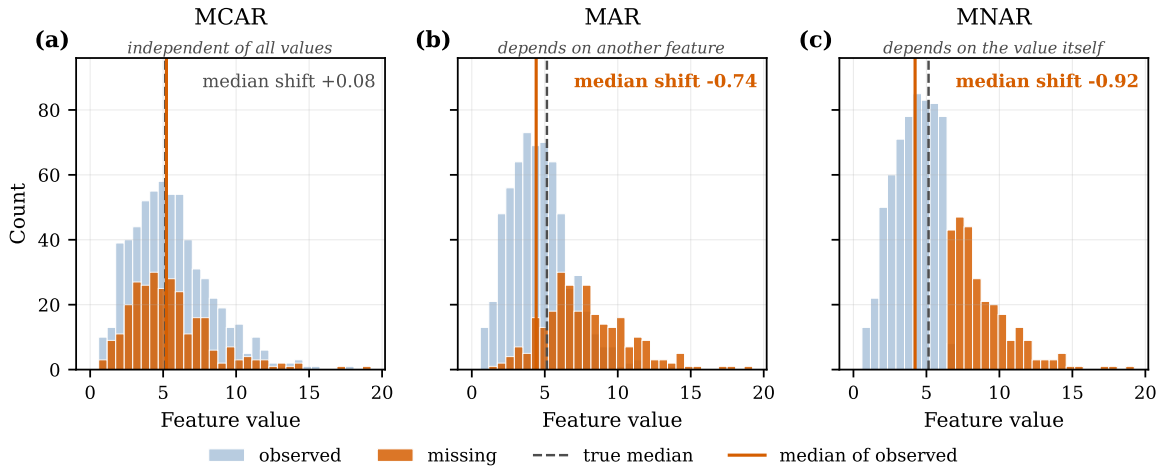


Figure 2.1: The same simulated feature with 30 % of its values removed under (a) MCAR, (b) MAR and (c) MNAR. Blue marks the values that remain observed, orange those removed. The dashed grey line is the median of the complete data, the solid orange line the median of the observed values, and the shift between them is stated in each panel. Figure adapted from [22].

This taxonomy structures the robustness experiments in Section 3.5.5 and predicts their outcome. The finding reported in Section 4.5, namely that MNAR degrades performance far more than MCAR for most pipelines, is the empirical signature of this theory.

The pipelines in this work use unconditional imputation: the median for numeric features and the most frequent category for categorical ones. This is the default in applied practice, it is what a generic pipeline would do, and it provides a common baseline against which the more elaborate pipelines add their own operations. Columns exceeding 50 % missing values

are dropped, on the grounds that the imputed value would then dominate the observed signal.

2.3.2 Outlier Treatment

Extreme values distort variance-based operations, and standardization, principal component analysis and any distance-based method are all sensitive to a small number of far-out points. Tukey [24] introduced the interquartile-range criterion that remains the standard non-parametric definition. With Q_1 and Q_3 the first and third quartile and $\text{IQR} = Q_3 - Q_1$, a value is considered an outlier if it falls outside

$$[Q_1 - k \cdot \text{IQR}; Q_3 + k \cdot \text{IQR}], \quad k = 1.5. \quad (2.1)$$

Two responses are available. *Trimming* removes the affected observations, which discards information and, in an imbalanced problem, may disproportionately remove minority-class instances. *Winsorization* clips the values to the boundary, retaining the observation while limiting its leverage. This work uses winsorization with bounds estimated on the training data only.

One implementation detail deserves emphasis because it proved consequential. Applied indiscriminately to all numeric columns, the criterion destroys binary indicator variables. For a one-hot dummy whose positive class occupies less than 25 % or more than 75 % of the rows, both quartiles coincide, the interquartile range is zero, and the winsorization interval collapses to a single point, so the entire column becomes constant. Outlier treatment is meaningful only for continuous features, and Section 3.3.2 documents the effect of ignoring this.

2.3.3 Feature Scaling

Numeric features in business data routinely differ by orders of magnitude: an age in years alongside a monetary turnover alongside an index value. Algorithms whose objective depends on the magnitude of feature values (regularized linear models, distance-based methods, gradient-descent-trained neural networks) are affected directly, because a feature with larger numerical range dominates both the gradient and the penalty term [25].

Three transformations are in common use. *Standardization* centers each feature on zero mean and scales it to unit variance. *Min-max scaling* maps to a fixed interval, typically $[0, 1]$. *Robust scaling* uses the median and interquartile range instead of mean and standard deviation, which reduces sensitivity to extreme values. This work uses standardization throughout, both because it is the most common default and because it is the prerequisite for principal component analysis (which operates on the covariance structure and therefore on centered, comparably scaled input).

Fig. 2.2 shows the operation on two predictors of the Bank Marketing dataset. In Fig. 2.2(a) both are drawn on a common horizontal axis: **age** occupies the range from 17 to 98 years while **euribor3m**, an interest rate in percent, is compressed into the leftmost sixth of the panel. A regularized linear model applied to this input would attribute most of its penalty to the age coefficient for no reason other than the unit in which age happens to be measured. After standardization, Fig. 2.2(b), both distributions are centered on zero with unit variance and occupy the same range, so neither can dominate through its unit alone. Their shapes are

unchanged, since standardization is a linear rescaling, so the interest rate remains concentrated on a few quoted values and age remains right-skewed.

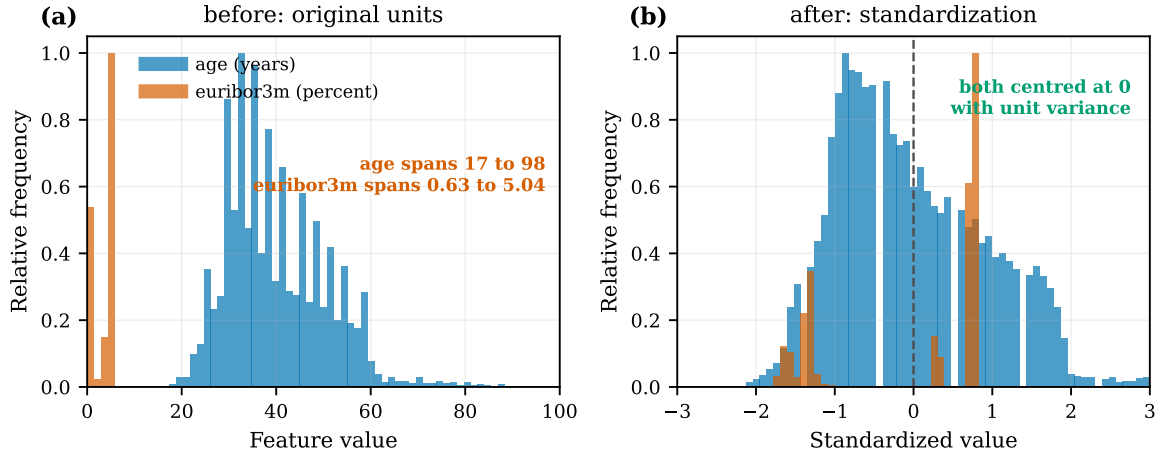


Figure 2.2: `age` and `euribor3m` of the Bank Marketing dataset (a) in their original units and (b) after standardization. Each histogram is normalized, so the two remain comparable despite very different bar counts. The dashed line in (b) marks zero.

Tree-based models are invariant to monotone rescaling of individual features, since their splits depend only on the ordering. That scaling nevertheless emerges in the ablation study of Section 4.4 as the most valuable of the elementary operations, surpassed only by the two most elaborate steps of the entire study, reflects the fact that the ablation uses logistic regression as its downstream proxy, a point taken up in the limitations.

2.3.4 Categorical Encoding

Categorical variables must be mapped to a numeric representation. *Label encoding* assigns an integer to each level, which is compact but imposes an artificial ordering that non-tree models will interpret as meaningful. *One-hot encoding* creates one binary indicator per level, avoiding the spurious ordering at the cost of dimensionality growth proportional to cardinality [25].

Two practical issues arise. First, categories present in the validation data but absent from the training data must be handled without introducing new columns. The convention adopted here aligns the validation set to the training columns and encodes unseen levels as all-zero. Second, one-hot encoding produces features whose scale differs from that of the raw numeric variables, which interacts with the scaling step. The ablation in Section 4.4 shows that one-hot encoding applied *without* subsequent standardization slightly degrades performance, illustrating that pre-processing steps are not independently beneficial.

A third option, *entity embeddings* [26], learns a dense low-dimensional vector per category as part of model training. This approach is the bridge to representation learning and is used inside Pipeline F.

2.3.5 Power Transformations

Many business quantities are strongly right-skewed: revenue, transaction volume, purchase frequency. Skewness affects linear models, whose error assumptions are symmetric, and it

interacts with scaling, since a single long tail inflates the standard deviation and compresses the bulk of the distribution.

Box and Cox [27] introduced the parametric family

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \ln y, & \lambda = 0, \end{cases} \quad (2.2)$$

with λ estimated by maximum likelihood (a profile likelihood over a grid of candidate values) so as to make the transformed variable as close to normally distributed as possible. The transformation requires strictly positive input (a restriction that matters on business data, where zero counts and credits are common), which limits where it can be applied.

Fig. 2.3 applies the transformation to the *Monetary* feature of Online Retail, the most strongly skewed quantity in either dataset. Fig. 2.3(a) requires a logarithmic count axis to remain readable at all, since almost every customer sits in the first bin while single customers reach two hundred thousand currency units. At the estimated λ the transformed distribution in Fig. 2.3(b) is close to symmetric, and the sample skewness falls from 18.1 to -0.01 .

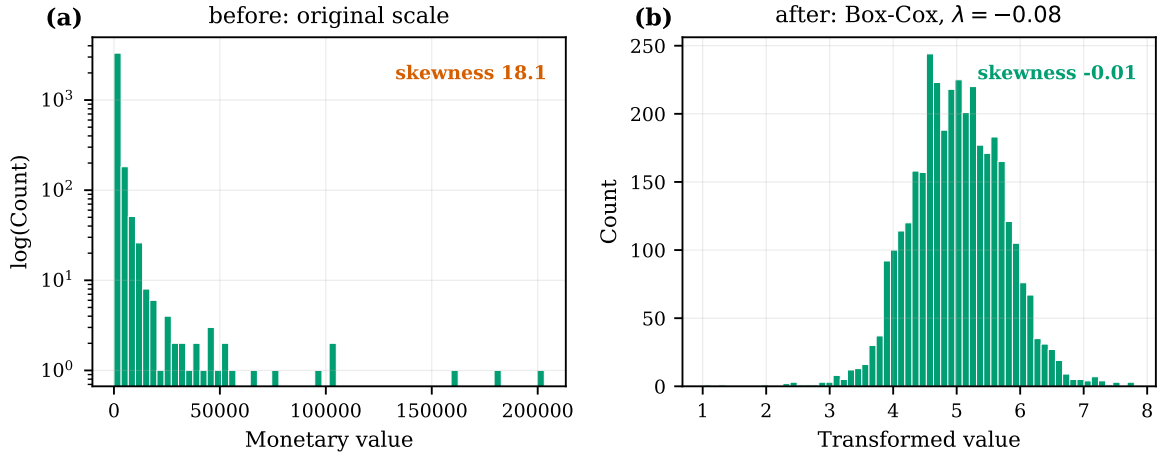


Figure 2.3: The *Monetary* feature of Online Retail (a) on its original scale with a logarithmic count axis and (b) after a Box-Cox transformation at the maximum-likelihood λ . The sample skewness is stated in each panel.

Because the transformation can produce values of very large magnitude when λ is estimated from a heavy-tailed sample, numerical safeguards are required in implementation. The pipelines here clip the transformed values before standardization.

2.3.6 Principal Component Analysis

Principal component analysis re-expresses correlated features in an orthogonal basis ordered by explained variance [25]. Retaining the leading components provides a lower-dimensional representation that preserves a specified share of the total variance (90 % and 20 % in this work).

PCA is unsupervised, which is the first of the two properties that matter here. It maximizes retained variance rather than class separability, so a direction that discriminates well but varies little may be discarded. The second is that the variance threshold is a consequential hyper-parameter rather than a technicality. This work therefore evaluates two settings, 90 % and 20 %, and treats the difference between them as a result.

Fig. 2.4 illustrates the three steps on the RFM features of Online Retail. Fig. 2.4(a) shows two correlated inputs with the two principal axes drawn in, the longer one following the direction of greatest variance. Fig. 2.4(b) shows the same points expressed in that basis, where the correlation has been removed and the second coordinate carries visibly less spread than the first. Fig. 2.4(c) reports the variance spectrum of all four features: the first component alone accounts for 73 % and two components pass the 90 % threshold, which is what makes a reduction from four dimensions to two defensible on this dataset.

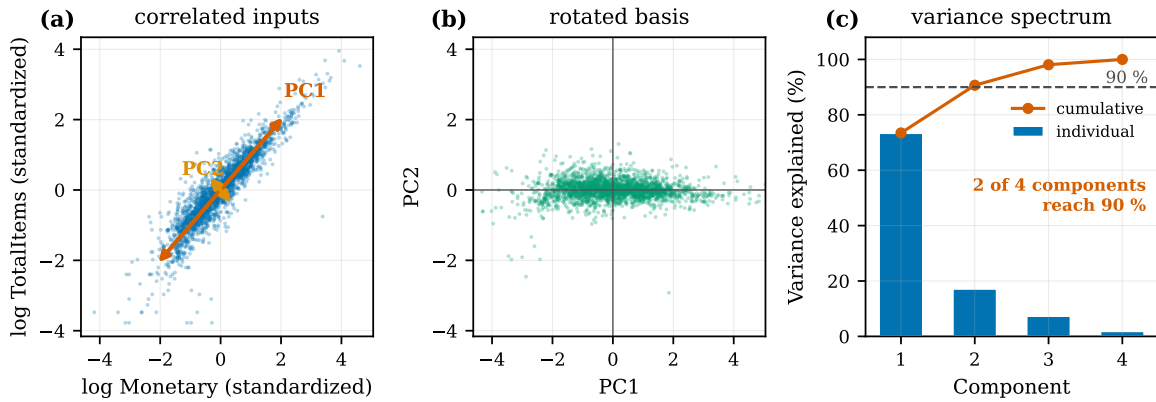


Figure 2.4: Principal component analysis on the log-transformed RFM features of Online Retail. (a) Two correlated inputs with the principal axes of that plane. (b) The same points in the rotated basis. (c) Variance explained by each of the four components, individually as bars and cumulatively as the line, with the 90 % threshold dashed. Panels (a) and (b) show the two-dimensional case, since a principal axis of the four-dimensional problem projected onto two of its dimensions is no longer orthogonal to the next.

PCA also has a property that becomes visible only in the robustness experiments: by projecting onto the dominant variance directions it discards the directions in which injected noise predominantly lies, which confers a passive robustness, provided the input contains redundant directions to discard in the first place.

2.3.7 Domain Feature Engineering

The operations discussed so far are generic, in that they apply to any tabular dataset without knowledge of its meaning. Domain feature engineering is the opposite, since it constructs features from subject-matter knowledge about what drives the outcome.

This work uses two mechanisms. *Interaction terms* form products of numeric features, allowing a linear model to represent multiplicative effects it could not otherwise capture. *Domain indicators and ratios* encode conventions specific to the dataset, such as decoding a sentinel value into an explicit binary flag, or expressing contact frequency relative to elapsed time.

For customer data the established framework is RFM analysis. Introduced in direct-marketing

practice [28] and given a theoretical foundation by Fader et al. [29], it characterizes a customer by three quantities: *recency* (the time since the last transaction), *frequency* (the number of transactions) and *monetary value* (the cumulative revenue). Wei et al. [30] review its applications. RFM matters for the present purpose because it is the kind of transformation that a learned representation would have to discover for itself, aggregating a variable-length transaction history into a fixed-length behavioral summary using domain reasoning about what constitutes customer value.

2.4 AI-based Representation Learning

The three architectures of Pipelines D to F share the idea that a useful feature space can be learned rather than constructed. They differ in their training objective, and that difference turns out to matter more for the results than the architecture itself.

2.4.1 Concept

Bengio et al. [9] frame representation learning as the task of discovering transformations of the data that make subsequent learning easier. The motivation is that the performance of machine learning depends heavily on how data is represented, and that constructing good representations by hand is laborious and requires domain expertise, exactly the cost that domain feature engineering incurs.

The promise for Business Intelligence is therefore direct, because if a representation can be learned from the data, the dependence on scarce subject-matter knowledge is reduced and the resulting pipeline may transfer between problems. Whether that promise holds for tabular business data is the question this work addresses.

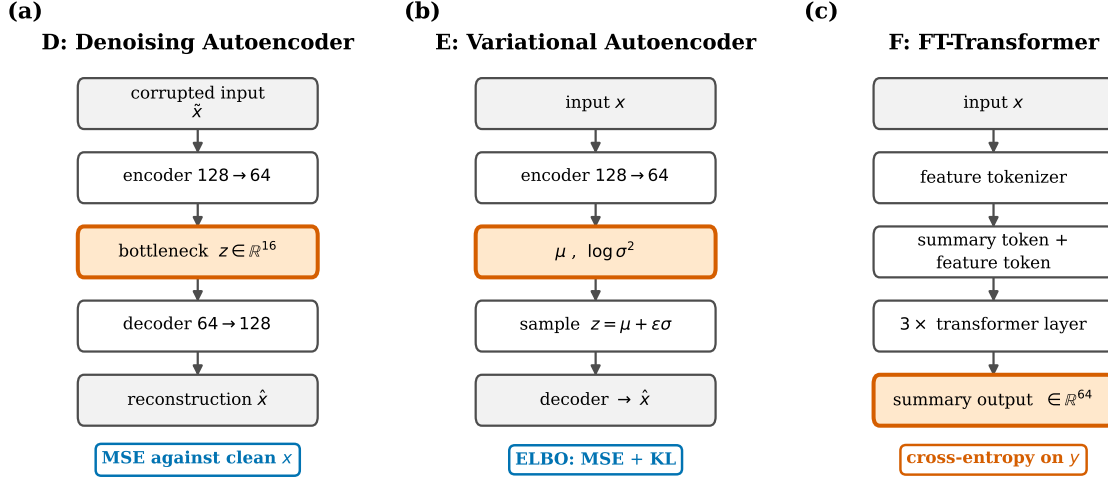
An important distinction concerns how a learned representation is used. It may serve as an intermediate layer inside an end-to-end classifier, in which case the representation and the decision function are trained jointly, or it may be extracted and passed to a separate downstream model. This work adopts the second formulation exclusively, because only that isolates the contribution of the representation, and because it corresponds to how a BI team with an established modeling stack would realistically adopt such a technique.

Fig. 2.5 sets the three architectures used here side by side. The denoising autoencoder in Fig. 2.5(a) and the variational autoencoder in Fig. 2.5(b) are trained to reconstruct their input and never see a label, whereas the FT-Transformer in Fig. 2.5(c) is trained against the class label and its representation is therefore aligned with the downstream task by construction. The three subsections that follow describe each in turn, and the difference in objective visible in the figure turns out to govern both the accuracy ordering and the robustness ordering reported in Chapter 4.

2.4.2 Autoencoders

An autoencoder consists of an encoder f_θ mapping the input to a latent code and a decoder g_ϕ mapping the code back to the input space, trained to minimize a reconstruction loss [31]:

$$\mathcal{L}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^n \|x_i - g_\phi(f_\theta(x_i))\|^2. \quad (2.3)$$



The orange frame marks the layer whose output is handed to the downstream classifier.
 D and E reconstruct their input; F discriminates the classes. This difference governs the robustness results.

Figure 2.5: Layer structure of (a) the denoising autoencoder of Pipeline D, (b) the variational autoencoder of Pipeline E and (c) the FT-Transformer of Pipeline F. The orange frame marks the layer whose output is passed to the downstream classifier. The training objective is stated below each column.

When the latent dimension is smaller than the input dimension, the network is forced to compress, and the bottleneck activation is used as the learned representation. With linear activations and squared error the optimum spans the same subspace as principal component analysis, while non-linear activations permit representations that PCA cannot express.

The objective has a consequence that becomes important in Section 4.5, since it rewards faithful reconstruction of the *training* distribution, not class discrimination. The latent space is therefore only meaningful for inputs resembling those the encoder was fitted on, which is the property that governs its behavior under corrupted input.

2.4.3 Denoising Autoencoder

Vincent et al. [10] modify the objective so that the network reconstructs a clean input from a deliberately corrupted version. Writing $\tilde{x}$ for the corrupted input,

$$\mathcal{L}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^n \|x_i - g_{\phi}(f_{\theta}(\tilde{x}_i))\|^2. \quad (2.4)$$

The network cannot succeed by learning the identity function and must instead capture dependencies between features, since only those permit a masked or noisy value to be recovered. Corruption is applied here as additive Gaussian noise combined with random masking of individual inputs.

The denoising objective is often described as producing robust representations. It is worth stating what kind of robustness is expected, because the encoder is trained to be invariant to the *specific corruption process* used during training. Whether this transfers to other corruption

mechanisms is an empirical question, and Section 4.5 answers it in the negative for systematic distortions such as MNAR missingness.

2.4.4 Variational Autoencoder

Kingma and Welling [11] modify the autoencoder in one respect. Where the encoder of Section 2.4.2 maps each record to a single point in the latent space, the encoder of a variational autoencoder maps it to a small region, outputting a centre μ and a width σ , and the code that the decoder receives is drawn at random from that region. Because neighboring records are then mapped to overlapping regions, the latent space has no gaps in which the decoder was never trained.

Training balances two demands, which are written together as the loss

$$\mathcal{L} = \underbrace{\|x - \hat{x}\|^2}_{\text{reconstruction}} + \underbrace{\beta D_{\text{KL}}}_{\text{regularization}} . \quad (2.5)$$

The first term is the ordinary reconstruction error and pulls the regions apart, because well-separated codes are easier to decode. The second term is the Kullback–Leibler divergence between the region the encoder produced and a standard normal distribution. It pulls the regions back together and towards the origin, so that the latent space stays compact. The weight β sets the balance and is left at one, its standard value, throughout this work.

Drawing a random sample would normally block gradient descent, since one cannot differentiate through a random draw. The reparameterization trick avoids this by drawing the randomness separately, as $z = \mu + \varepsilon\sigma$ with ε from a standard normal distribution. The random part no longer depends on the network weights, so gradients pass through μ and σ unimpeded.

The regularization term is useful when the goal is to generate new records, but for feature extraction it is a constraint that may discard exactly the information a classifier would have used, since compactness of the latent space and separability of the classes are unrelated objectives. And at inference no sampling takes place, since the centre μ is used directly as the representation, which makes the output deterministic.

2.4.5 Feature Tokenizer Transformer

The transformer architecture [32] replaced recurrence with self-attention and became the dominant approach in sequence modeling. Adapting it to tabular data requires a way of turning a fixed set of heterogeneous features into a token sequence.

Gorishniy et al. [12] propose the Feature Tokenizer Transformer. Each feature is embedded into a d -dimensional token: numeric features through a learned affine projection $t_j = x_j \cdot w_j + b_j$, categorical features through embedding lookup in the manner of entity embeddings [26]. An additional token is prepended that belongs to no feature and is learned during training. Its purpose is to act as a collector: because the attention mechanism lets every token read every other, this summary token accumulates information from the whole record while the feature tokens remain tied to their own column. The sequence passes through several transformer encoder layers, and the final state of the summary token serves as the encoding of the whole

record. TabTransformer [13] follows a related design in which only categorical features are tokenized.

What distinguishes this architecture from the autoencoders, and what proves relevant to the results, is a pair of properties. Self-attention makes the contribution of each feature dependent on the others, which allows a degraded feature to be down-weighted relative to the remainder. More consequentially, the architecture is trained *supervised*, with a classification head. Its representation is therefore optimized for class discrimination rather than for reconstruction, which helps performance and robustness, but which creates a serious evaluation hazard discussed in Section 2.7.

2.5 Downstream Classification Models

Because the object of study is the representation, the downstream models are held constant across pipelines. Five are used, spanning the range from a regularized linear model to gradient-boosted ensembles and a neural network, so that the question of whether a pre-processing effect is model-specific can be answered.

2.5.1 Logistic Regression

Logistic regression models the log-odds of the positive class as a linear function of the features [25]. Regularization via an L_1 , L_2 or elastic-net penalty controls overfitting and, in the L_1 case, enables implicit feature selection.

Its role here is twofold: as one of the five downstream models it represents the linear, fully interpretable end of the spectrum, while as the proxy model in the ablation it supplies a deterministic response to changes in the representation. It is deterministic because, unlike the ensembles, it involves no randomness in fitting, which matters when the effect being measured is small. Because it is sensitive to feature scaling and cannot represent interactions unless they are supplied explicitly, it responds particularly clearly to pre-processing.

2.5.2 Random Forest

A random forest aggregates many decision trees, each fitted to a bootstrap sample and considering a random subset of features at each split [25]. Individual trees are grown deep and therefore have low bias but high variance. Averaging over trees that have been decorrelated twice, once by the bootstrap and once by the random feature subset, reduces that variance considerably while leaving the bias largely intact.

Three properties matter for the present comparison. The forest is invariant to monotone transformations of individual features, since a split depends only on the ordering of values and not on their magnitude, so scaling and Box-Cox leave it unaffected in principle. It handles mixed feature types without explicit encoding decisions. And it requires comparatively little tuning to reach reasonable performance, which makes it a fair opponent even under the modest budget used here. The first of these properties is what the ablation study tests directly, since a pipeline effect that survives a scale-invariant learner cannot be a scaling effect.

2.5.3 Gradient Boosting: XGBoost and LightGBM

Gradient boosting builds an additive ensemble in which each new tree fits the gradient of the loss with respect to the current prediction [25]. Two implementations dominate applied use, of which XGBoost [33] adds explicit regularization of tree complexity, second-order gradient information and column subsampling. LightGBM [34] uses histogram-based split finding and leaf-wise tree growth, which reduces training time considerably on larger datasets.

Both are the reference methods for tabular prediction, and their strength is the reason the comparative literature discussed in Section 2.8 finds deep architectures struggling to gain ground on tabular problems. Grinsztajn et al. [7] trace that strength to three inductive biases, and all three are visible in the results of this work. Boosted trees tolerate uninformative features, because a feature that never reduces the loss is simply never selected for a split, which matters when Pipeline C expands the input to 144 columns. They are invariant to monotone rescaling for the same reason as the random forest. And they represent irregular, non-smooth decision boundaries without difficulty, whereas a neural network is biased towards smooth functions and must expend capacity to approximate a step. Tabular targets are frequently of exactly that irregular kind.

2.5.4 Multi-Layer Perceptron

The multi-layer perceptron is a feed-forward neural network [31]. Each layer computes a weighted sum of the layer below it, adds a bias and applies a non-linear activation. Without that non-linearity the whole stack would collapse into a single linear map, so it is the activation that lets the network represent feature interactions a linear model would have to be given explicitly. The final layer produces one value that is mapped to a probability.

Its capacity is set by the width of a layer (32 to 512 units) and the depth (one to four layers), both tuned here. Training uses the Adam optimizer [35], with dropout and weight decay counteracting overfitting. Implementation uses PyTorch [36].

The perceptron is included because its inductive bias resembles that of the encoders in Pipelines D to F, since both learn distributed representations by gradient descent. If a learned representation were to suit one downstream model in particular, this would be that model, so its results check whether the pipeline ordering is an artifact of using tree-based learners.

One implementation detail carries into the results. The perceptron is the only downstream model here that weights its loss by the inverse class ratio, which shifts its decision threshold towards the minority class. PR-AUC is threshold-independent and therefore unaffected, but the perceptron reaches the highest F_1 and the worst Brier scores of all five models. Section 4.10 returns to this.

2.6 Evaluation Methodology

Comparing seven pipelines across five downstream models requires an evaluation protocol that is both fair and sensitive enough to resolve small differences. This section covers the metrics, the resampling scheme, the tuning budget and the statistical apparatus, together with the two properties of repeated cross-validation that limit what the significance tests can establish.

2.6.1 Metrics for Imbalanced Classification

For a binary classifier with a tunable threshold, two threshold-independent summaries are in common use. The *receiver operating characteristic* (ROC) plots the true positive rate against the false positive rate, while the *precision–recall curve* (PR) plots precision against recall.

Davis and Goadrich [14] show that a curve dominating in ROC space also dominates in PR space and conversely, but that the two summaries respond very differently to imbalance. The false positive rate has the number of true negatives in its denominator. When negatives are abundant, a large absolute number of false positives changes it only slightly, and ROC-Area-under-curve (AUC) remains high. Precision has the number of predicted positives in its denominator and reacts directly. Saito and Rehmsmeier [15] confirm empirically that the PR curve is the more informative summary for imbalanced data.

The area under the precision–recall curve summarizes that curve in a single number between zero and one. Unlike ROC-AUC, whose uninformative baseline is always 0.5, the baseline of PR-AUC equals the positive rate of the dataset, because a classifier ranking at random attains a precision equal to that rate at every recall. The values reported in this work must therefore be read against 0.113 on Bank Marketing and 0.508 on Online Retail. A PR-AUC of 0.473 on Bank Marketing is roughly four times the random baseline, while the visually larger 0.780 on Online Retail is only about 1.5 times its own. Absolute PR-AUC values are not comparable between datasets for this reason, and only the ordering within a dataset carries meaning.

For this reason PR-AUC is the primary metric throughout this work. It is reported alongside:

- **ROC-AUC**, for comparability with the literature
- F_1 , **precision and recall** at the default threshold, which depend on that threshold and are therefore secondary
- **Brier score**, the mean squared error of the predicted probabilities, which measures calibration rather than discrimination and exposes the MLP behavior noted above
- **Lift@Top-10 %**, the ratio of the positive rate in the highest-scoring decile to the overall base rate, the metric that translates model quality into campaign yield

The last is included because a purely statistical comparison does not answer the question a BI decision-maker asks. A PR-AUC difference of 0.013 is uninterpretable in isolation, while the same difference expressed as a change in campaign yield is not.

2.6.2 Cross-Validation

A single train–test split provides an estimate whose variance depends on which observations happened to fall into which partition. Kohavi [37] established k -fold cross-validation as the standard remedy, in which the data is partitioned into k folds, each serving once as the validation set. *Stratified* folds preserve the class ratio, which is essential under imbalance, and *repeated* cross-validation averages over several partitionings to reduce the dependence on any one.

This work uses 5 folds with 3 repetitions, giving 15 scores per configuration. The choice trades

off between the stability of the estimate and the computational cost, which is multiplied by the number of configurations (70 in the main experiment).

2.6.3 Hyper-parameter Optimization

Comparing pre-processing pipelines requires that each be given a fair opportunity to perform, which means the downstream model must be tuned for each. Bergstra and Bengio [38] showed that random search over hyper-parameters outperforms grid search at equal budget, because grid search wastes evaluations on dimensions that do not matter. Bayesian approaches improve further by modeling the relationship between hyper-parameters and objective.

Optuna [39] implements a tree-structured Parzen estimator, which models the densities of well- and poorly-performing configurations and samples where their ratio is favorable. This work uses 20 trials per pipeline-model combination with PR-AUC as the objective.

The critical design requirement is that the budget be *identical* across configurations. An unequal budget would confound the comparison, because a pipeline that appeared worse might simply have received less tuning.

2.6.4 Statistical Significance

Observed performance differences may reflect fold-to-fold variation rather than a genuine effect. Demšar [16] sets out the methodology for classifier comparison and recommends non-parametric tests, since the assumptions of the paired t -test (normality and commensurability of differences) are typically violated by cross-validation scores. His procedure is designed for comparisons *across a collection of datasets*, in which each dataset supplies one independent observation. That scope condition is essential and is easily overlooked when the same test is applied to the folds of a single dataset, as Section 4.3 discusses for this study.

The *Wilcoxon signed-rank test* [40] ranks the absolute differences between paired observations and compares the rank sums of positive and negative differences. It requires only symmetry of the difference distribution.

With many pairwise comparisons the probability of at least one false positive grows rapidly. The *Bonferroni correction* [41] divides the significance level by the number of comparisons, controlling the family-wise error rate at the cost of statistical power. With 35 configurations per dataset, 595 pairwise comparisons arise and the corrected level becomes $\alpha_{\text{corr}} \approx 8.4 \times 10^{-5}$.

Two properties of this apparatus limit its interpretation, both encountered in Section 4.3. The Wilcoxon statistic on 15 paired samples is discrete, with a minimum attainable p -value of $2/2^{15} \approx 6.1 \times 10^{-5}$. Many comparisons attain this floor, at which point the test can confirm that a difference exists but cannot rank differences by magnitude. More seriously, the scores of a *repeated* cross-validation are not independent, because the repetitions re-use the same data and the folds within a repetition share most of their training material. Nadeau and Bengio [42] quantify this and derive a corrected resampled t -test whose variance estimate is inflated by the ratio of test-set to training-set size. Uncorrected procedures understate the variance and reject too readily. Both properties point the same way: the magnitude of a difference is read off the difference in PR-AUC itself rather than from a standardized statistic.

2.6.5 Ablation Studies

An aggregate performance figure for a pipeline consisting of several operations does not reveal which operation is responsible. Ablation, meaning the systematic removal or isolation of components, attributes the effect.

Two designs are possible. *Leave-one-out* removes a single step from the complete pipeline and measures the loss, which quantifies each step’s marginal contribution in the presence of the others. *Isolated application* applies each step alone to the raw data, which quantifies its standalone contribution. This work uses the second design, extended by the complete pipeline for comparison, because the interest is in whether the elaborate operations carry the performance or whether elementary ones suffice.

A composition of operations need not be additive, since steps interact and a combination may perform worse than its best constituent. Detecting such interaction is one purpose of comparing the isolated steps against the complete pipeline.

2.7 Information Leakage

Kaufman et al. [43] provide the formal treatment on which this section rests. They define leakage as the introduction of information about the target that should not legitimately be available for inference, and they make *legitimacy* the central concept: a quantity is legitimate for predicting a target if it is observable to the client at the moment the prediction is required. For predictive modeling this reduces to what they call the no-time-machine requirement, namely that a model may use only information observed no later than the target it predicts. Their remedy is a *learn-predict separation*, a cut through the data that admits as features only what was legitimate at prediction time, and as training examples only what was legitimate with respect to every evaluation target.

Fig. 2.6 draws that separation as a timeline and marks the three ways in which this work encountered a violation of it. Each orange arrow is information moving backwards across the cut, which is the graphical form of a broken no-time-machine requirement.

The consequence is an evaluation estimate that overstates deployed performance, often considerably, and the error is insidious because nothing in the results appears anomalous. Kapoor and Narayanan [44] show how widespread the problem has become, surveying 294 affected papers across 17 fields and setting out a taxonomy of eight leakage types. The three forms treated below correspond to their types L2 (illegitimate features), L3.1 (temporal leakage) and L1.2 (pre-processing on train-test data). They were selected from the eight because these are the ones that arose here.

2.7.1 Post-hoc Features

A feature recorded only after the predicted event has occurred cannot be used to predict it. Such features are frequently highly predictive because they are consequences of the outcome, which makes them attractive and easy to retain inadvertently. Identifying them requires reasoning about the temporal order of data generation rather than inspection of the data itself.

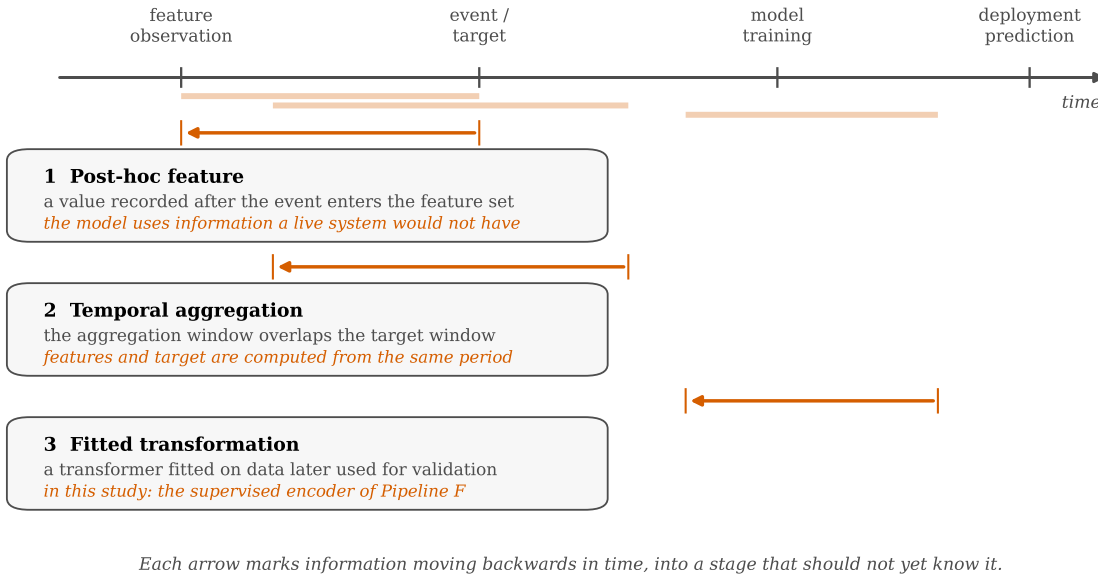


Figure 2.6: The three leakage mechanisms relevant to this work, placed on the timeline of a modeling task. Each orange arrow runs backwards from the stage holding the information to the stage that should not yet possess it, and the shaded bar above marks the affected span. Figure adapted from [43].

2.7.2 Temporal Leakage in Aggregated Features

When features aggregate a history and the target is measured over a period, aggregation windows that overlap the target period import information about the target. The remedy is a temporal split: features from strictly before a cutoff, target from strictly after.

2.7.3 Leakage Through Fitted Transformations

Any transformation whose parameters are estimated from data (an imputation statistic, a scaler, a PCA basis, a learned encoder) must be fitted on the training portion only. Fitting on the full dataset before splitting transfers information from the validation partition into the training procedure.

Kaufman et al. [43] treat this case under their learn-predict separation without grading it, and the gradation that follows is a derivation of the present work. The severity of this last form scales with the capacity of the transformation. For a scaler, which estimates two parameters per feature, the effect is negligible. For a learned encoder with many thousands of parameters, it is not. The most severe case arises when the transformation is fitted *using the labels*, because a supervised encoder trained on the full training set can encode label information about individual observations into their representations. A downstream model cross-validated on those representations then has access, through the features, to the labels of its own validation folds.

This configuration is not exotic. It is the natural way to use a supervised architecture as a feature extractor, namely training the encoder, exporting the representation and evaluating downstream models on it. Kapoor and Narayanan [44] survey 294 affected papers across 17 disciplines and provide a taxonomy of eight leakage types, several of which arise from fitted

transformations of this kind. Section 3.6 documents the magnitude of the resulting artifact in this study and the procedure used to eliminate it.

2.8 Related Work and Positioning

The question whether deep architectures can compete with tree ensembles on tabular data has been studied repeatedly, with a consistent answer. What has not been studied is the question this work asks, and the gap becomes visible once the framing of the existing studies is made explicit.

2.8.1 Deep Learning on Tabular Data

Shwartz-Ziv and Armon [8] evaluate several proposed deep tabular models on datasets outside those used in their original publications and find that XGBoost outperforms them, that no deep model is consistently best across datasets, and that an ensemble of deep models and XGBoost outperforms either alone. They also note that the deep models required substantially more tuning effort to reach their reported performance.

Grinsztajn et al. [7] go further and investigate the causes on a curated benchmark. They identify three inductive biases favoring tree ensembles on tabular data: robustness to uninformative features, invariance to monotone feature transformations, and the ability to represent irregular, non-smooth target functions. Neural networks, by contrast, are biased toward smooth solutions and are affected by uninformative features and by feature scaling.

Borisov et al. [45] survey the field and reach a similar conclusion while noting that deep approaches retain specific advantages: they integrate naturally with other modalities, support transfer and self-supervised pre-training, and permit end-to-end learning in larger systems. Wang et al. [46] review the self-supervised branch of that literature in detail.

The picture has continued to move. McElfresh et al. [6] quantify the conditions under which either family wins, Gorishniy et al. [47] show that a parameter-efficient MLP ensemble outperforms attention- and retrieval-based tabular architectures, and Hollmann et al. [48] report a transformer pre-trained on synthetic data that surpasses tuned gradient boosting on datasets of up to roughly ten thousand samples. The last of these is relevant to the present work in that it achieves its result through large-scale pre-training rather than through a representation learned on the target dataset alone, which is the setting examined here.

Architecturally, the relevant proposals include TabNet [17], which uses sequential attention for instance-wise feature selection, TabTransformer [13], which applies self-attention to categorical embeddings, and the FT-Transformer [12], which tokenizes all features and which the present study employs.

2.8.2 The Gap Addressed Here

The studies above share a framing: the deep architecture is evaluated as a *classifier*, competing against a tree ensemble on the same input. Three consequences follow.

First, the comparison confounds the quality of the learned representation with the quality of the classification head. A deep model may lose because its representation is uninformative,

or because its final layer is a weaker decision function than a boosted ensemble. The design cannot distinguish these.

Second, the framing does not correspond to how such techniques would enter a BI organization. A team with an established gradient boosting stack does not replace it wholesale, while it might well feed learned features into it. That usage, representation learning as a pre-processing operator, is the practically relevant configuration and is not what the literature evaluates.

Third, the classical side of the comparison is typically a single fixed pre-processing configuration, reported briefly if at all. Since the same literature establishes that data preparation strongly influences outcomes, holding it fixed and unreported leaves open whether the observed differences reflect the architectures or the preparation.

This work addresses all three. Every pipeline, classical and learned, is used strictly as a pre-processing operator whose output is passed to the same five downstream models, which isolates the representation. The classical side is decomposed into three pipelines of increasing elaboration and further into individual operations by ablation, which makes the pre-processing effect visible rather than assumed. And the comparison is extended beyond accuracy to robustness and interpretability, which are the criteria that determine adoption in a Business Intelligence context.

Chapter 3

Methodology and Experimental Setup

This chapter describes the experimental apparatus used to isolate the effect of pre-processing on downstream predictive performance. The design follows a factorial layout in which every pre-processing pipeline is combined with every downstream classifier on every dataset, so that the pipeline effect can be read independently of the model choice. Section 3.1 introduces the two datasets, Section 3.2 the factorial design, Sections 3.3 and 3.4 the six pipelines, and Section 3.5 the evaluation protocol. Section 3.6 documents three sources of information leakage that were identified and eliminated during development. They are described in detail because each of them materially changed the results.

3.1 Datasets

Two publicly available business datasets were selected. They differ deliberately in structure, size and target definition, which allows the robustness of the findings across different Business Intelligence scenarios to be assessed.

3.1.1 Bank Marketing

The UCI Bank Marketing dataset [49] records 41,188 contacts from a Portuguese banking institution’s direct marketing campaigns. The binary target indicates whether the contacted client subscribed to a term deposit. With 11.3% positive cases the dataset is strongly imbalanced, which motivates PR-AUC as the primary metric (Section 3.5.1). The dataset contains no missing values.

After removing the target the dataset provides 20 predictors, ten numeric and ten categorical. Two properties required explicit treatment.

The duration feature is excluded. `duration` records the length of the last contact in seconds and is therefore only known *after* the call has ended. A model using it could not be deployed for campaign targeting, and the dataset documentation explicitly warns that it should be discarded for realistic predictive modeling. Retaining it inflates PR-AUC considerably. All pipelines in this work operate on the remaining 19 predictors.

The value `pdays = 999` is a sentinel, not a quantity. `pdays` encodes the number of days since the client was last contacted in a previous campaign, with 999 denoting “not previously contacted”. This applies to 96.3% of all rows, while the genuine values range from 0 to 27 days. Treated as a number, the feature has a mean of 962 and a median of 999, which distorts scaling, correlation structure and any subsequent principal component analysis. Only Pipeline C decodes the sentinel into a binary indicator. This asymmetry is intentional: recognizing and encoding such domain-specific conventions is the capability that Pipeline C is meant to represent, and leaving it unhandled elsewhere reflects what a generic pre-processing pipeline would actually do.

3.1.2 Online Retail

The UCI Online Retail dataset [50] contains 541,909 transaction records of a UK-based online retailer between December 2010 and December 2011. Cleaning removes cancellations (9,288 records), non-positive quantities (10,624), non-positive unit prices (2,517) and transactions without a customer identifier (135,080), leaving 397,884 records for 4,338 customers across 37 countries. These counts overlap and must not be added: every cancellation carries a negative quantity and is therefore contained in the second category, and records lacking a customer identifier occur in all of them. The number removed in total is 144,025.

Transactions are aggregated to customer level using the RFM framework [28, 29, 30], yielding four features: *Recency* (days since the last purchase), *Frequency* (number of distinct invoices), *Monetary* (total revenue) and *TotalItems* (total quantity purchased).

Target definition and temporal split. The target is defined as customer retention, so features are aggregated strictly from transactions before a cutoff date, and the target records whether the customer transacted again in the subsequent observation window. This deviates from the original research proposal, which envisaged a binary high-value label. A high-value label derived from monetary turnover would be a deterministic function of the *Monetary* input feature and would therefore constitute target leakage. The temporal retention formulation avoids this and additionally reflects the way churn models are deployed in practice. After aggregation 3,616 customers remain in the modeling dataset.

3.2 Experimental Design

The experiment is a full factorial design over three factors:

- **Pre-processing pipeline** (7 levels): A, B_{PCA90}, B_{PCA20}, C, D, E, F
- **Downstream classifier** (5 levels): Logistic Regression, Random Forest, XGBoost, LightGBM, Multi-Layer Perceptron
- **Dataset** (2 levels): Bank Marketing, Online Retail

Fig. 3.1 shows the resulting grid for both datasets. The colouring separates the two ways a representation can be produced, since the blue cells of Fig. 3.1(a) and (b) use a transformation fitted once per fold, while the orange cells belong to Pipelines D, E and F, whose encoder is re-trained inside every fold for the reason given in Section 3.6.

$7 \times 5 \times 2 = 70$ configurations, 15 cross-validation splits each

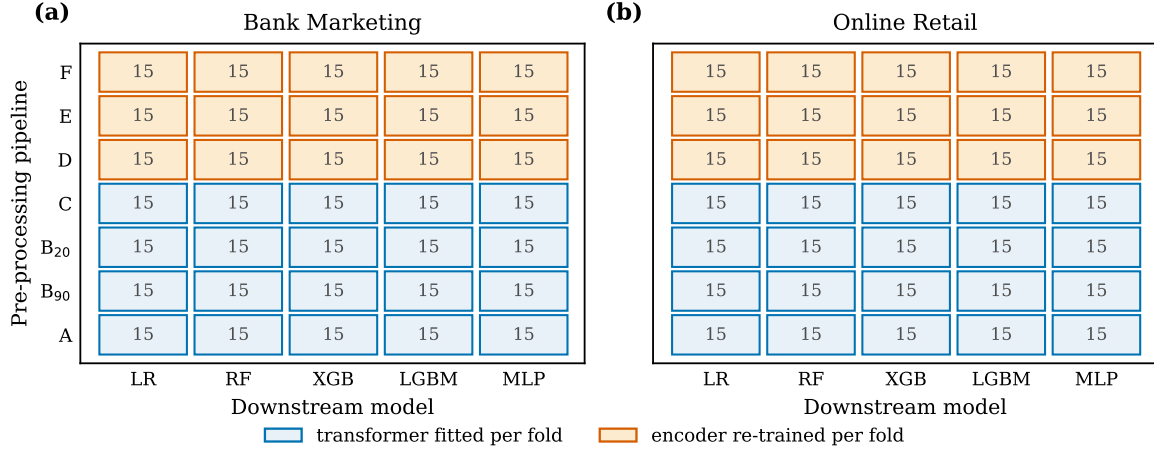


Figure 3.1: The seven pre-processing variants against the five downstream classifiers for (a) Bank Marketing and (b) Online Retail. The number in each cell is the count of cross-validation splits over which that configuration is evaluated. Blue cells use a transformation fitted once per fold, orange cells an encoder re-trained inside every fold.

This yields $7 \times 5 \times 2 = 70$ pipeline-model-dataset combinations, each evaluated over 15 cross-validation splits. Pipeline B is instantiated at two variance thresholds because the amount of variance retained by PCA is itself a pre-processing decision whose effect is of interest.

The breadth of the model factor is deliberate. If the pre-processing effect were observed for a single classifier only, it could not be separated from an interaction between that classifier and the representation. Evaluating all five models allows the question of whether the pipeline effect is model-independent or model-specific to be addressed directly.

3.3 Classical Pre-processing Pipelines

All classical pipelines fit their transformers on the training portion only and apply them to the validation portion without refitting. The descriptions that follow give each pipeline at the level of its operations. Appendix A.5 states those operations that carry a design decision, together with the encoder fitting protocol and the corruption mechanisms, precisely enough to be re-implemented without the repository. Fig. 3.2 sets the seven variants side by side. Before the individual descriptions, two things are worth reading off it. All seven variants share the same first two operations, so the pipelines differ in fewer steps than their descriptions suggest, and the output dimensionality at the right-hand end spans from four to 144 features on the same input, which is the range within which the results of Chapter 4 have to be interpreted.

3.3.1 Pipeline A: Baseline

The baseline applies the three operations that are near-universal in applied practice, namely median imputation for numeric features and most-frequent imputation for categorical ones [22, 23], one-hot encoding of categorical variables [25], and standardization of all resulting columns. Columns exceeding 50% missing values are dropped. On Bank Marketing this produces 62 features from 19 predictors.

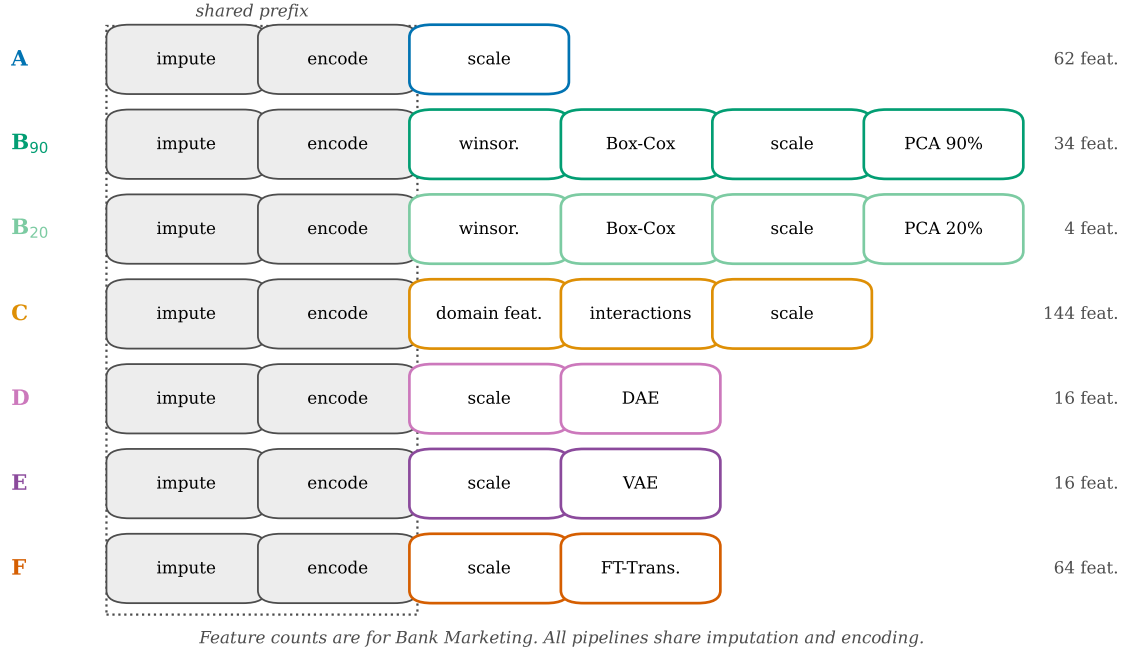


Figure 3.2: The operation sequence of every pipeline, aligned so that shared steps occupy the same column. The grey steps inside the dotted rectangle are common to all seven variants. The number of features produced on Bank Marketing is stated at the right-hand end of each chain.

Pipeline A serves as the reference against which the added value of every more elaborate pipeline is measured, and it is the configuration a practitioner would reach for without further consideration.

3.3.2 Pipeline B: Statistical Extension

Pipeline B extends the baseline with three classical statistical corrections: IQR-based winsorization of outliers at $k = 1.5$ [24], a Box-Cox power transformation to reduce skewness [27], and principal component analysis at a variance threshold of either 90 % (B_{PCA90}) or 20 % (B_{PCA20}).

Winsorization is restricted to continuous features. Applying IQR winsorization indiscriminately to all numeric columns destroys one-hot indicator variables, because for a dummy whose positive class occupies less than 25 % or more than 75 % of the rows, the interquartile range is zero, the winsorization bounds collapse to a single point, and the entire column becomes constant. On Bank Marketing this affected 46 of 62 columns and reduced the number of components retained by PCA at 90 % variance from 34 to 9. Winsorization is therefore applied only to columns with more than two distinct values.

3.3.3 Pipeline C: Domain Feature Engineering

Pipeline C adds domain-specific feature construction on top of the baseline operations. For Bank Marketing this comprises indicator and ratio features derived from campaign semantics (among them the decoding of the `pdays` sentinel into a binary *previously contacted* flag and a contact-intensity ratio), followed by polynomial interaction terms between numeric features.

For Online Retail the RFM aggregation itself constitutes the domain feature engineering, again extended by interaction terms. On Bank Marketing the pipeline produces 144 features.

Pipeline C represents the manual, knowledge-driven end of the spectrum and is the direct classical counterpart to the learned representations of Pipelines D–F.

3.4 AI-based Representation Learning Pipelines

The three AI pipelines share a common protocol. Each is used as a *pre-processing operator*, not as an end-to-end classifier, so an encoder is trained on the training data and the resulting latent representation is passed to the same five downstream classifiers used for the classical pipelines. This is what makes the comparison meaningful, since the only thing that varies is the representation handed to the classifier.

The encoders were implemented in PyTorch[36], with Adam optimization and a batch size of 256.

3.4.1 Pipeline D: Denoising Autoencoder

The denoising autoencoder [10] compresses the input through a $128 \rightarrow 64 \rightarrow 16$ encoder and reconstructs it through a mirrored decoder. During training the input is corrupted with additive Gaussian noise ($\sigma = 0.2$) and random masking ($p = 0.3$), while the reconstruction target remains the clean input. The 16-dimensional bottleneck activation is used as the representation.

3.4.2 Pipeline E: Variational Autoencoder

The variational autoencoder [11] replaces the deterministic bottleneck with a probabilistic one, so the encoder outputs a mean μ and a log-variance $\log \sigma^2$, and the latent code is sampled using the reparameterization trick $z = \mu + \varepsilon \cdot \sigma$. Training optimizes the evidence lower bound, combining reconstruction error with the Kullback–Leibler divergence to a standard normal prior ($\beta = 1$). At inference the deterministic mean μ is used as the representation, again with 16 dimensions.

3.4.3 Pipeline F: FT-Transformer

The Feature Tokenizer Transformer [12] embeds every feature into a token, numeric features through a learned linear projection and categorical features through embedding lookup [26]. A learned summary token is prepended, as described in Section 2.4.5, and the sequence is processed by a three-layer transformer encoder with eight attention heads and pre-layer normalization [32]. The final state of that summary token, 64-dimensional, serves as the extracted feature vector.

Unlike D and E, this encoder is trained *supervised*, with a cross-entropy classification head. This is faithful to the original architecture but has an important consequence for evaluation, discussed in Section 3.6.

Early stopping on a validation split. All three encoders use early stopping monitored on a held-out validation split (15% of the training data), not on the training loss. Monitoring the training loss provides no generalization signal, since the encoder simply continues

until it has memorized the training data. For the supervised FT-Transformer this is particularly damaging, since memorized label information is written directly into the extracted representation.

3.5 Evaluation Protocol

The protocol is identical for every pipeline, which is what allows the pre-processing effect to be read from the results. It fixes the resampling scheme, the metrics, the tuning budget and the statistical tests, and adds two supplementary experiments that the aggregate figures alone would not support.

3.5.1 Cross-Validation and Metrics

Every combination is evaluated with 5×3 repeated stratified k -fold cross-validation [37], giving 15 scores per combination. Stratification preserves the class ratio in every fold. A held-out test split (20 %) is reported alongside the cross-validation estimate as an independent check.

The primary metric is the area under the precision–recall curve (PR-AUC). For imbalanced problems, ROC-AUC is dominated by the abundant negative class and provides optimistic values that do not reflect performance on the minority class of interest [14, 15]. Secondary metrics comprise ROC-AUC, F_1 , precision, recall and the Brier score. One business-oriented metric is added to connect statistical performance to decision value: Lift@Top-10 %, the factor by which the positive rate in the highest-scoring decile exceeds the base rate.

3.5.2 Hyper-parameter Optimization

Each downstream model is tuned with Optuna [39] using a tree-structured Parzen estimator, 20 trials, and PR-AUC evaluated by an internal three-fold cross-validation on the training split as the objective [38]. The same budget applies to every pipeline–model combination, so that no pipeline is advantaged by receiving more tuning effort.

Deviation from the proposal. The research proposal specified nested cross-validation. The implemented scheme performs hyper-parameter selection on the training split and reports the outer 5×3 cross-validation, which is not fully nested. Fully nested optimization would multiply runtime by the number of outer folds and was not feasible within the available compute budget. The consequence is a mild optimistic bias in the absolute cross-validation estimates. It does not compromise the *comparison*, which is the object of this study, because the bias applies uniformly to all combinations. The held-out test split provides an unbiased reference.

3.5.3 Statistical Significance

Pairwise comparisons use the Wilcoxon signed-rank test [40] on the 15 cross-validation fold scores, following the methodology established by Demšar [16] for classifier comparison. With 35 pipeline–model combinations per dataset, 595 pairwise comparisons result, and the significance level is Bonferroni-corrected [41] to $\alpha_{\text{corr}} = 0.05/595 \approx 8.4 \times 10^{-5}$. The magnitude of a difference is reported as the difference in mean PR-AUC together with the fold standard deviation, both in the unit of the metric itself.

This choice turned out to be the wrong one, and Section 4.3 sets out why. The 15 scores

of a repeated cross-validation are not independent, which the signed-rank test assumes. The corrected resampled t -test of Nadeau and Bengio [42] would have been the appropriate instrument. The test results are reported in the following chapter for completeness but are not used to support any conclusion. All comparisons in this work are decided on effect sizes, that is on differences in mean PR-AUC read against the fold standard deviation, which is why the design retains its value despite the flawed inference procedure.

3.5.4 Ablation Study

To attribute performance to individual pre-processing operations, each pipeline is decomposed into three stages, the raw input without pre-processing, each constituent step applied in isolation to the raw data, and the complete pipeline. Comparing the isolated steps identifies which operation carries the effect, while comparing the isolated steps against the complete pipeline reveals whether the combination produces synergy. Logistic regression is used as a fast and deterministic downstream proxy throughout the ablation.

3.5.5 Robustness Tests

Robustness is assessed by injecting synthetic data quality problems into the training fold only, leaving the validation fold clean. Five mechanisms are applied at increasing intensity: missing values completely at random (MCAR), at random (MAR) and not at random (MNAR) following the taxonomy of Little and Rubin [22], outlier injection, and label noise. Missing-data rates range from 5 % to 50 %.

For Pipelines D–F the encoder trained in the main experiment is applied unchanged to the corrupted data. This measures the *inference-time* robustness of a deployed representation rather than the robustness of the training procedure, a distinction that must be kept in mind when interpreting the results. Where corruption pushes a feature past the 50 % missing-value threshold, the column is re-inserted as missing rather than dropped, because the input dimensionality of a deployed encoder is fixed and cannot change at inference time.

3.6 Prevention of Information Leakage

Three distinct leakage mechanisms were identified during development. Each of them, left uncorrected, would have produced a materially different and misleading answer to the research questions. They are documented here because the correction of the third is itself a methodological contribution of this work.

3.6.1 Post-hoc Features

As described in Section 3.1.1, the `duration` feature of the Bank Marketing dataset is only observable after the outcome-determining event and is excluded from all pipelines.

3.6.2 Temporal Leakage in Aggregated Features

RFM features aggregated over the entire observation period would incorporate information from the period in which the target is measured. The Online Retail pipeline therefore aggregates strictly before a cutoff date and observes the target strictly afterwards.

3.6.3 Encoder Leakage in Representation Learning

The most consequential mechanism concerns the AI pipelines. The natural implementation, which fits the encoder once on the training data, exports the representation and cross-validates a downstream model on it, is invalid. Every validation fold in the cross-validation was part of the data the encoder was fitted on. For the unsupervised autoencoders this constitutes transductive leakage and proves to be minor in practice. For the *supervised* FT-Transformer it is severe, because the encoder has seen the validation samples *together with their labels* and writes that information into the extracted embedding.

The magnitude is considerable. On Bank Marketing, the naive protocol provides a PR-AUC of 0.858 for Pipeline F, against a true value of approximately 0.457. Three independent estimates confirm the corrected figure: the held-out test split (0.359), the leakage-free ablation with per-fold encoder training (0.373 at reduced training budget), and a controlled experiment in which an identical model is evaluated on held-out rows the encoder had seen versus rows it had not (0.820 versus 0.391, a ratio of 2.10). The corresponding ratio for the unsupervised pipelines D and E lies between 0.93 and 0.99, confirming that they are unaffected.

Uncorrected, this artifact would have supported the conclusion that representation learning clearly outperforms classical pre-processing, the opposite of the finding reported in Chapter 4.

The correction re-fits the encoder inside every cross-validation fold, using only that fold’s training portion. Since the fold partition is deterministic and the encoder does not depend on the downstream model, the 15 fold encodings are computed once and cached for reuse across all five classifiers. For reasons of compute budget the per-fold encoder training is capped at 60 epochs, against 100 for the single-pass pipeline runs.

3.7 Reproducibility

All randomness is controlled by a single seed (42) applied to NumPy, PyTorch and all scikit-learn splits [51]. Every pipeline exports its fitted transformers and output dimensionality. The complete implementation, including the experiment orchestration script that reproduces all results end to end, is published together with this work.

Chapter 4

Results and Discussion

This chapter reports the empirical results and interprets them with respect to the two research questions. Section 4.1 characterizes the two datasets. Section 4.2 presents the predictive performance of all 70 pipeline–model–dataset combinations and answers RQ1, supported by the significance analysis in Section 4.3 and the ablation study in Section 4.4. Section 4.5 addresses RQ2. Sections 4.6 and 4.7 add the business and interpretability perspectives that motivate the study in the first place. Section 4.9 synthesizes the findings and Section 4.10 states their limits.

4.1 Exploratory Data Analysis

The two datasets differ in size, feature count and target definition, and those differences account for much of what follows. This section records the characteristics that recur in the interpretation of the results.

4.1.1 Bank Marketing

The dataset comprises 41,188 records with 19 predictors after exclusion of the post-hoc **duration** feature. The target is subscription to a term deposit, positive in 4,640 cases, corresponding to a positive rate of 11.3%. There are no missing values, so imputation is inactive on this dataset in the undisturbed setting, while it becomes relevant in the robustness experiments of Section 4.5. Fig. 4.1 shows the class distribution. A ratio of roughly eight to one is moderate by the standards of fraud or failure detection but severe enough that accuracy is uninformative, since a model predicting the negative class throughout would already score 88.7%.

The campaign structure leaves two further marks on the data. Contacts are concentrated in a few months, with May alone accounting for 13,769 of 41,188 records, so any month indicator carries campaign timing rather than seasonality of customer behavior. And **campaign**, the number of contacts made in the current campaign, is itself a consequence of the sales process rather than a property of the client. It reaches 56 in the extreme while the median is two, and its upper tail describes clients who were called repeatedly without success.

The numeric features divide into two groups. Client attributes (**age**, **campaign**) are mod-

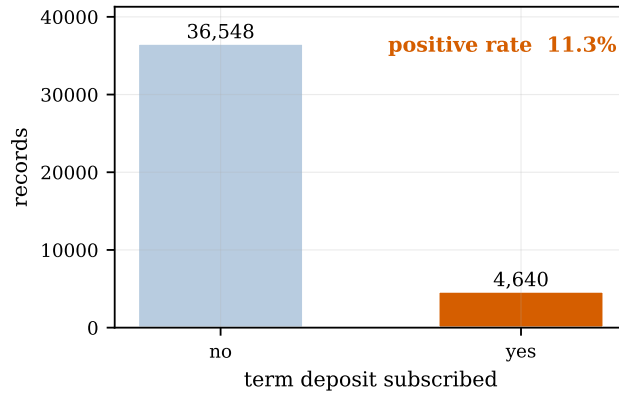


Figure 4.1: Class distribution of the Bank Marketing target: 36,548 contacts without a subscription against 4,640 with one.

erately skewed and behave conventionally. Macroeconomic indicators (`emp.var.rate`, `cons.price.idx`, `euribor3m`, `nr.employed`) vary on entirely different scales and are strongly correlated with one another, reflecting the fact that they describe the same economic cycle. This correlation structure is directly relevant to Pipeline B, because it is what allows principal component analysis to compress the feature space substantially, and it is equally the reason why the resulting components are dominated by macroeconomic variation rather than by client characteristics.

Fig. 4.2 shows the nine numeric predictors individually. The client attributes in Fig. 4.2(a), (b) and (d) are unimodal and right-skewed. The five macroeconomic indicators in Fig. 4.2(e) to (i) are visibly multimodal, because each is constant within a campaign period and the dataset spans only a handful of such periods. Fig. 4.2(c) required a logarithmic count axis because of the sentinel discussed below.

The categorical features are dominated by a small number of frequent levels, with `poutcome` taking the value *nonexistent* in 35,563 of 41,188 rows. As discussed in Section 3.1.1, the numeric feature `pdays` is a sentinel-coded variable in 96.3 % of rows.

The cardinality of the categorical block is low throughout, between two and twelve levels, which is what makes one-hot encoding viable at all, since the 19 predictors expand to 62 columns rather than to several hundred. Fig. 4.3 shows the correlation structure of the numeric block. The five macroeconomic indicators form a tightly coupled group with coefficients up to 0.97, while the client attributes are close to uncorrelated with everything else. This is the redundancy that principal component analysis in Pipeline B exploits, and it explains why its leading components are macroeconomic rather than client-specific.

4.1.2 Online Retail

Cleaning reduces the 541,909 raw transactions to 397,884 usable records covering 4,338 customers, 18,532 invoices and 3,665 products in 37 countries over the period December 2010 to December 2011. Transactions are heavily concentrated: the United Kingdom accounts for 354,321 of 397,884 records.

The RFM aggregation produces four strongly right-skewed features. Monetary value in par-

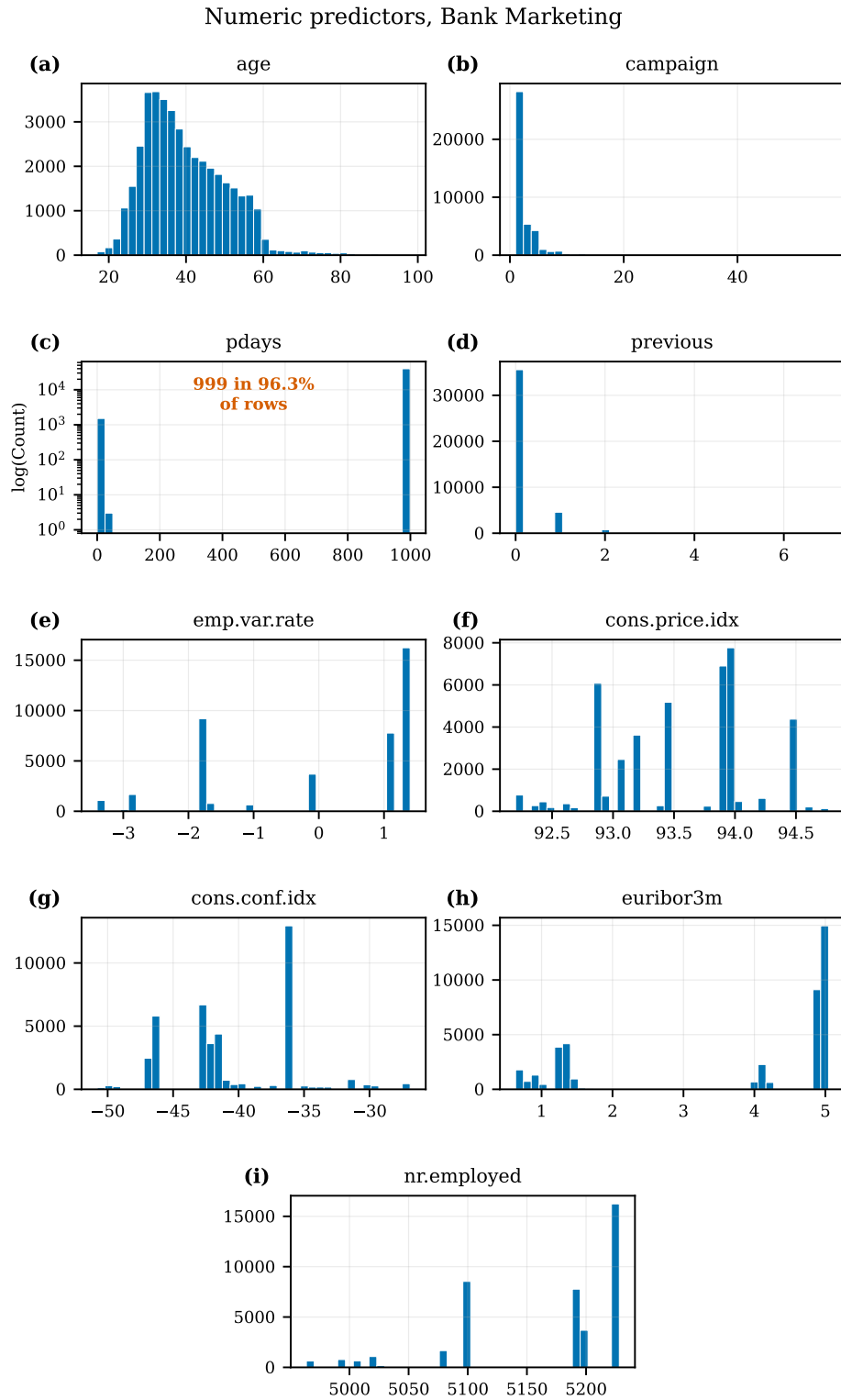


Figure 4.2: Histograms of the nine numeric predictors of Bank Marketing: (a) `age`, (b) `campaign`, (c) `pdays`, (d) `previous`, (e) `emp.var.rate`, (f) `cons.price.idx`, (g) `cons.conf.idx`, (h) `euribor3m` and (i) `nr.employed`. Panel (c) uses a logarithmic count axis.

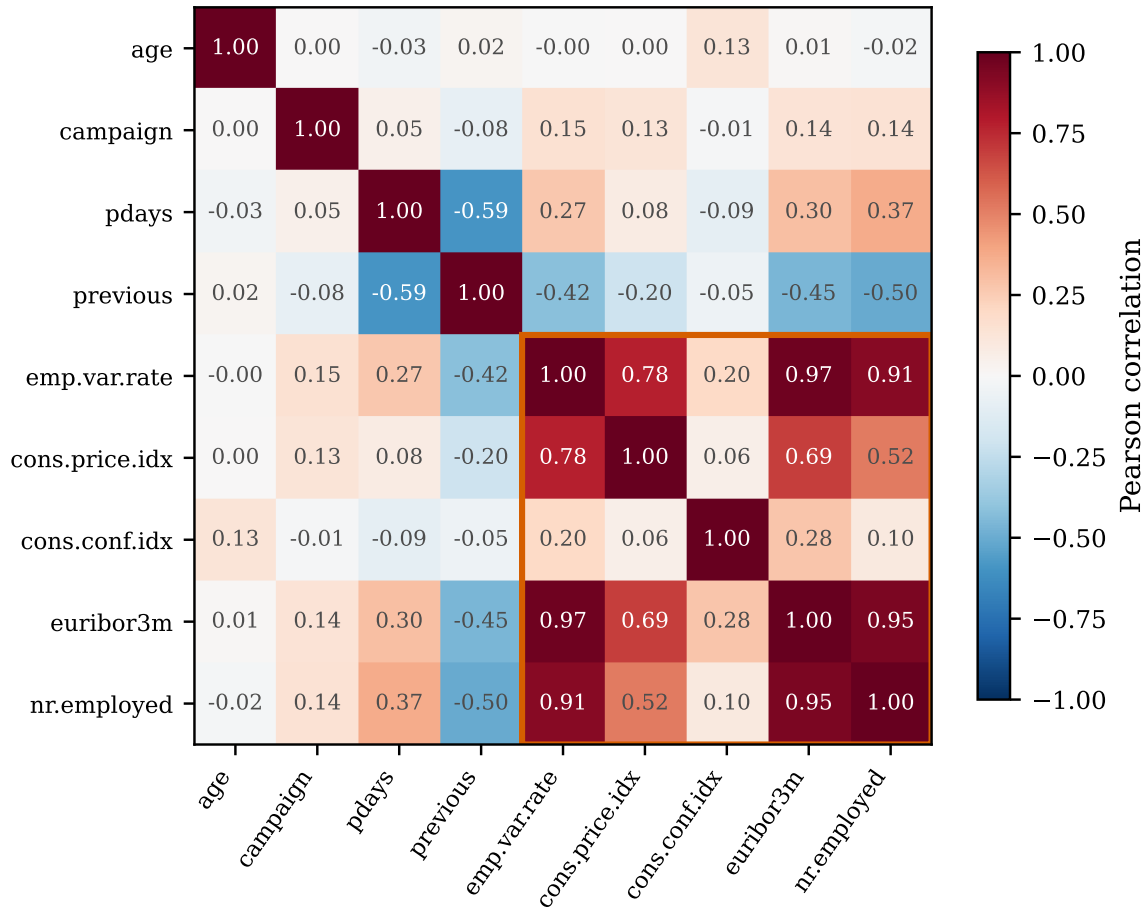


Figure 4.3: Pearson correlation matrix of the nine numeric predictors of Bank Marketing. Red indicates positive and blue negative correlation. The orange frame encloses the five macroeconomic indicators.

ticular spans four orders of magnitude, from 3.75 to 280,206 currency units, with a median of 674, these figures being computed over the entire cleaned observation period. The modelling features are aggregated only up to the cutoff date, so the corresponding values in Fig. 4.4 are somewhat smaller, at a median near 600 and a maximum above 200,000. This skew is what Pipeline B’s Box-Cox transformation is designed to address, and the ablation study confirms that this is the one dataset where that transformation pays off.

After aggregation only four predictors remain, so there is far less scope for pre-processing to add or destroy information than on Bank Marketing. The temporal split additionally reduces the modeling population to 3,616 customers, a small sample by the standards of representation learning. Both factors compress the differences between pipelines, as the following sections show.

Fig. 4.4 shows the four aggregates. All of them need a logarithmic count axis, and Fig. 4.4(c) and (d) illustrate the extent of the skew: a median around 600 currency units against a maximum above 200,000. Fig. 4.5 places the temporal split in context. Features are computed exclusively from the shaded period before the cutoff and the target from the period after it, which is what keeps the target from being a function of its own inputs.

RFM features after customer-level aggregation

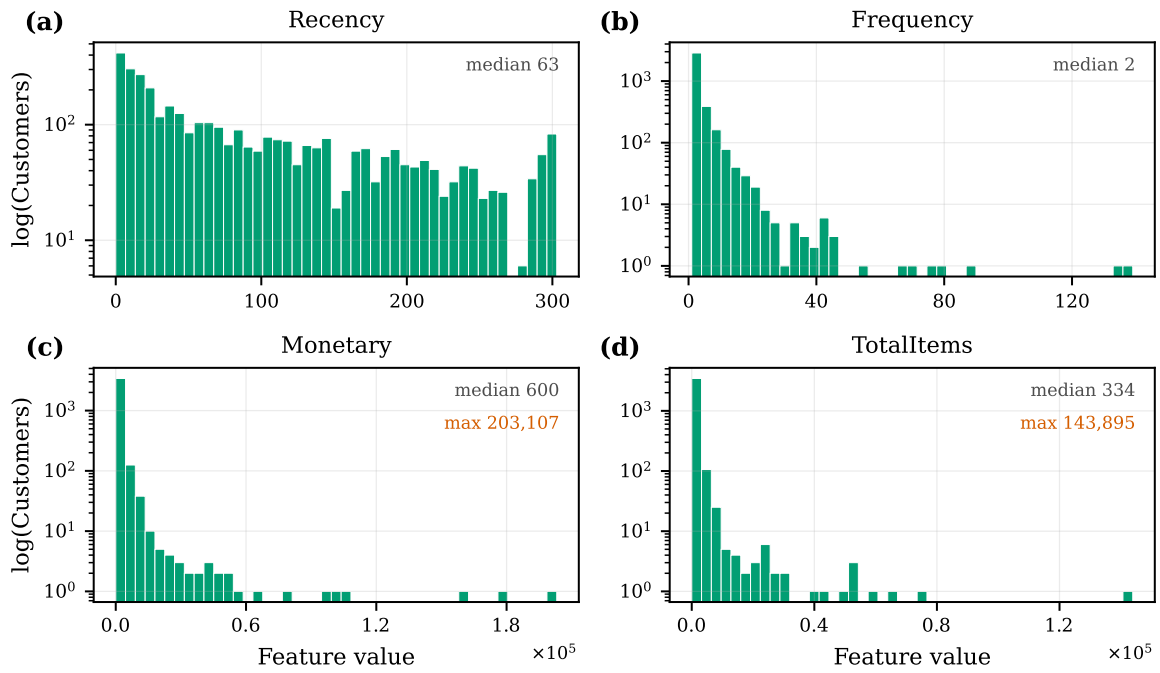


Figure 4.4: Distributions of (a) Recency, (b) Frequency, (c) Monetary and (d) TotalItems after aggregation to customer level over the feature window. All four count axes are logarithmic. Median and, for the two largest quantities, maximum are stated in each panel. The horizontal axes of (c) and (d) carry a shared factor of 10^5 .

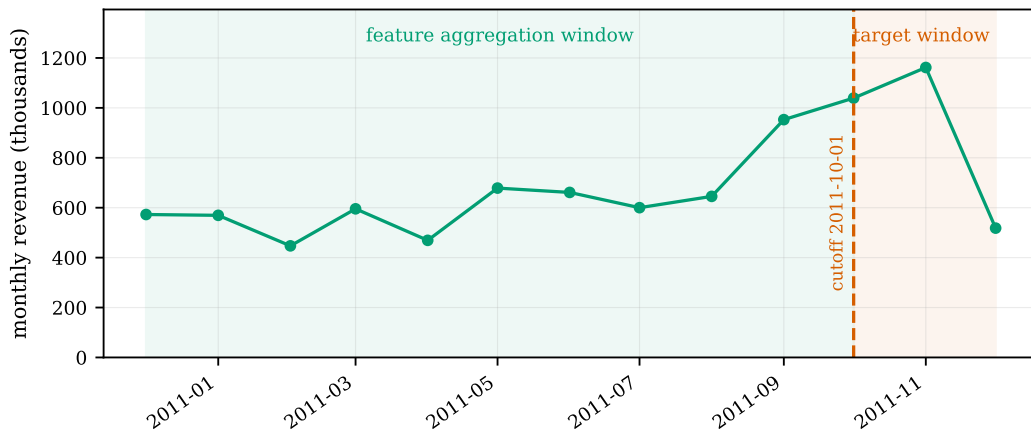


Figure 4.5: Monthly revenue of Online Retail over the observation period. The dashed line marks the cutoff of 1 October 2011, the green area the feature aggregation window and the orange area the target observation window.

Table 4.1: PR-AUC per pipeline as a cross-validation mean over 15 folds, read in two ways. The *best* column gives the strongest of the five downstream models, named in parentheses. The *mean* column averages over all five. The best-of-five reading favours pipelines that suit one learner particularly well, so both are given. Rows are ordered by the best column on Bank Marketing.

Pipeline	Bank Marketing		Online Retail	
	best	mean	best	mean
C: Feature Engineering	0.4728 (XGB)	0.4615	0.7797 (XGB)	0.7772
A: Baseline	0.4703 (LGBM)	0.4610	0.7721 (RF)	0.7681
F: FT-Transformer	0.4572 (XGB)	0.4554	0.7635 (XGB)	0.7627
E: VAE	0.4342 (RF)	0.4255	0.7623 (MLP)	0.7607
D: DAE	0.4309 (XGB)	0.4267	0.7641 (RF)	0.7626
B: PCA 90 %	0.4264 (XGB)	0.4220	0.7637 (LR)	0.7561
B: PCA 20 %	0.3285 (RF)	0.3126	0.7627 (LR)	0.7569

4.2 Predictive Performance

The 70 pipeline–model–dataset combinations answer RQ1 directly. The results are presented first in aggregate, then broken down by downstream model to establish whether the pipeline effect depends on the learner, and finally compared across the two datasets.

4.2.1 Overall Picture

Each of the seven pipelines was paired with each of the five downstream classifiers and evaluated on both datasets, so every pipeline is represented by five PR-AUC values per dataset rather than by a single number. The comparison below reports the best of those five, which is the fairest reading for a pre-processing operator, since a practitioner choosing a pipeline would also choose the classifier that works best with it. Restricting the comparison to one fixed classifier changes the individual values and, for two of the five classifiers, the pipeline at the top, as Section 4.2.2 shows. The best-of-five reading is therefore a selection rule rather than a neutral summary, and where it matters for a conclusion this is stated.

Reading Table 4.1 by the mean rather than by the best model leaves the top of the ordering intact and compresses the distances. On Bank Marketing the gap between Pipelines C and A falls from 0.0025 to 0.0006, and the only exchange further down is between D and E. On Online Retail the two leaders keep their places while the five pipelines behind them reshuffle, which is what a spread of 0.017 across seven pipelines predicts and which supports the reading given in Section 4.2.3 rather than contradicting it.

Table 4.1 summarizes the best downstream model per pipeline, while Figs. 4.6 and 4.7 show the complete 7×5 matrix per dataset, and the complete numerical results including all secondary metrics are given in Appendix A.3.

The values must be read against the positive rate of the respective dataset, which is where the PR-AUC of a random ranking lies. That baseline is 0.113 on Bank Marketing and 0.508 on Online Retail, so the smaller numbers on Bank Marketing represent the larger improvement over chance. Only the ordering within a dataset carries meaning.

The ordering is identical on both datasets in its essential feature: Pipeline C exhibits the high-

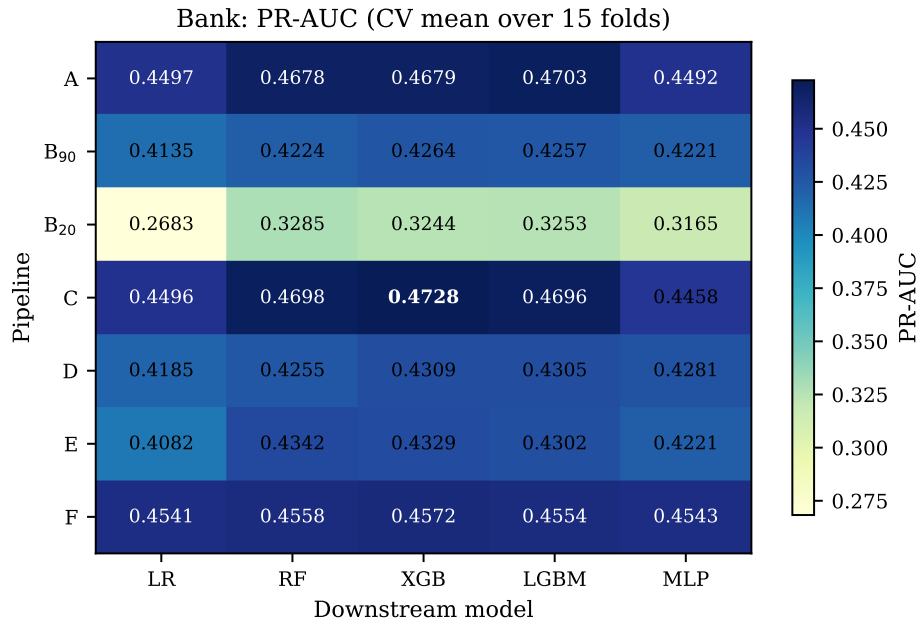


Figure 4.6: Cross-validation PR-AUC for every pipeline and model on Bank Marketing, averaged over 15 folds. Darker cells are higher and the best value is set in bold.

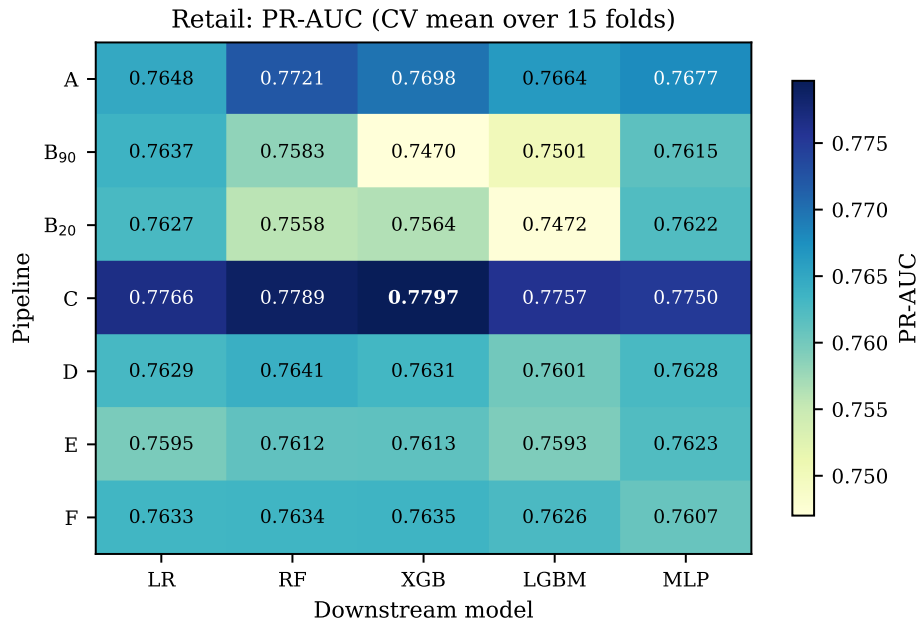


Figure 4.7: Cross-validation PR-AUC for every pipeline and model on Online Retail, averaged over 15 folds. Darker cells are higher and the best value is set in bold. Note that the colour range spans 0.033, against 0.204 in Fig. 4.6.

est PR-AUC, and under the best-of-five reading no AI-based pipeline surpasses the classical baseline. The qualification matters, since for two individual classifiers on Bank Marketing Pipeline F does edge past the baseline, as Section 4.2.2 sets out. The negative answer to RQ1 is the central finding of this work. The ordering of C against A within the classical family is a far weaker result and is qualified below.

Three observations qualify this result.

The margin between C and A is small. Read by the best downstream model, C exceeds A by 0.0025 PR-AUC on Bank Marketing and by 0.0076 on Online Retail. Read by the mean over the five models, the second figure becomes 0.0090. Which of the two readings is quoted is stated wherever the number appears, since they differ. Given fold standard deviations of 0.013 to 0.021 across the ten runs involved, both differences are well within one standard deviation. On Bank Marketing the lead additionally holds for only two of the five downstream models, so the comparison between C and A is left undecided on that dataset. The elaborate feature engineering of Pipeline C, which more than doubles the feature count from 62 to 144 on Bank Marketing, therefore buys an improvement that is at best marginal, and Section 4.4 shows that what little it buys comes from the interaction terms rather than from the domain features.

The spread within the classical family exceeds the spread between families. On Bank Marketing the range from B_{PCA20} (0.3285) to C (0.4728) is 0.144, whereas the entire AI family spans 0.4309 to 0.4572, a range of 0.026 that sits comfortably inside the classical range. The decisive question is therefore not whether a pipeline is classical or learned, but which specific operations it performs.

Aggressive dimensionality reduction is the only clearly harmful choice. B_{PCA20} loses 0.14 PR-AUC on Bank Marketing by compressing 62 features into four components. On Online Retail, where only four features exist to begin with, the same setting costs almost nothing (0.7627 versus 0.7721). Information destroyed by compression cannot be recovered by any downstream model, and the magnitude of the loss scales with how much information was there to destroy.

4.2.2 The Effect Is Largely Model-Independent

Across all five classifiers the pipeline ranking is broadly but not perfectly stable. On Bank Marketing, Pipeline C ranks first or second for three of five models, and B_{PCA20} ranks last for all five. The exceptions are logistic regression and the perceptron, where Pipeline F leads, so the ordering does change at the top for those two learners. The statement that no AI pipeline surpasses the classical baseline therefore holds for the best of the five downstream models rather than for every individual one. The margin that carries the RQ1 conclusion, 0.016 PR-AUC, is unaffected because it compares each pipeline at its own best classifier. The largest model-specific deviation concerns the MLP, which suffers a clearly worse Brier score while achieving comparable PR-AUC. Its values lie between 0.152 and 0.198 across the seven pipelines, against 0.077 to 0.088 for the three tree ensembles, which indicates poorly calibrated probability estimates. The per-combination figures are listed in Tables A.5 and A.6. Both metrics reported in this chapter, PR-AUC and lift, depend only on the ranking of the predicted scores and are therefore unaffected. The miscalibration would matter as soon as the predicted values were read as probabilities, as any expected-value calculation in deployment would do.

Its cause is a class weighting that is applied to the perceptron alone, which Section 4.10 takes up.

That the pipeline effect is largely independent of the downstream model strengthens the interpretation, since what is being measured is a property of the representation rather than an interaction between a particular representation and a particular learner.

4.2.3 Dataset Dependence

The absolute performance levels differ sharply between the datasets (PR-AUC of approximately 0.45 versus 0.77), but this reflects the different base rates and target definitions rather than a difference in pipeline quality. More informative is the *compression* of differences on Online Retail, where the spread between the best and worst pipeline is 0.144 on Bank Marketing but only 0.017 on Online Retail.

The explanation lies in the feature space. Online Retail provides four carefully constructed RFM aggregates that already encode the relevant customer behavior, so there is little for any pre-processing strategy to improve upon and little to destroy. Bank Marketing provides 19 raw predictors of mixed type and scale, leaving considerable room for pre-processing to matter. This suggests a practical rule: the value of sophisticated pre-processing scales with the rawness of the input.

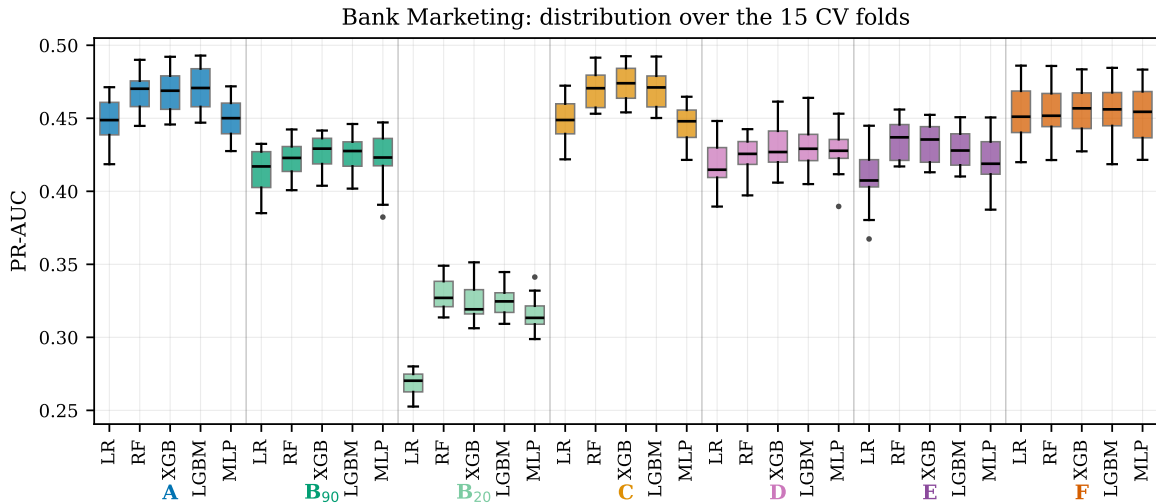


Figure 4.8: PR-AUC across the 15 cross-validation folds for all 35 pipeline and model combinations on Bank Marketing. Boxes span the interquartile range, the line marks the median and whiskers extend to 1.5 times that range. Colour identifies the pipeline.

Figs. 4.8 and 4.9 show these results as distributions rather than as means, and the compression is visible directly. On Online Retail the boxes overlap almost completely for every configuration except the two PCA variants, whereas on Bank Marketing the fold-to-fold spread of roughly 0.012 to 0.019 still exceeds several of the differences between neighbouring pipelines. That spread is the reason the comparison in Section 4.3 is carried out on paired folds rather than on the pipeline means.

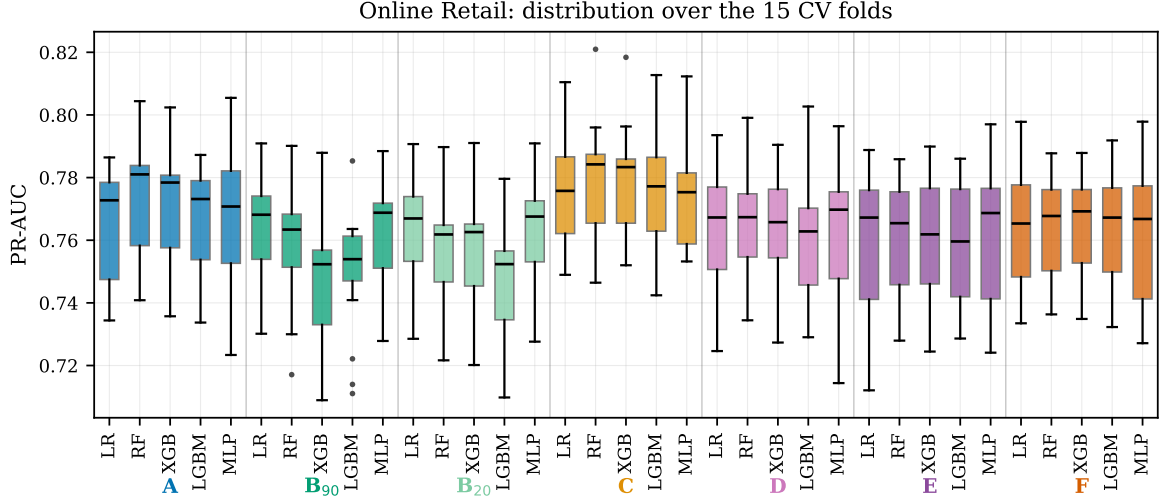


Figure 4.9: PR-AUC across the 15 cross-validation folds for all 35 pipeline and model combinations on Online Retail, drawn as in Fig. 4.8.

4.3 Statistical Significance

Of the 595 pairwise comparisons per dataset, 401 are nominally significant on Bank Marketing and 117 on Online Retail after Bonferroni correction ($\alpha_{\text{corr}} \approx 8.4 \times 10^{-5}$). Restricting attention to comparisons that cross the classical/AI boundary, 196 of 300 are significant on Bank Marketing and 43 of 300 on Online Retail. The difference between the datasets mirrors the compression described above, since where all pipelines perform within 0.017 PR-AUC of one another, few differences are detectable at all.

These counts are reported for completeness, but they should not be used to support any of the conclusions in this chapter, for two reasons.

The fold scores are not independent. The Wilcoxon signed-rank test assumes independent paired observations. The 15 scores of a 5×3 *repeated* cross-validation do not satisfy this. The three repetitions evaluate the same data under different partitions, and the five folds within a repetition share 80% of their training data with one another. The effective sample size is therefore considerably smaller than 15, the variance of the mean difference is underestimated, and the null hypothesis is rejected more often than the nominal level implies. The corrected resampled *t*-test of Nadeau and Bengio [42], which inflates the variance estimate by the ratio of test-set to training-set size to account for exactly this overlap, is the appropriate instrument for repeated cross-validation and was not used here. The Bonferroni correction does not repair this, since it addresses multiplicity rather than dependence. With an understated variance it merely offsets one error with another.

The methodology is cited outside its intended scope. Demšar [16] develops the signed-rank procedure for comparing classifiers *across multiple datasets*, where each dataset contributes one independent observation. Applying it to the folds of a single dataset is a different situation from the one the paper addresses, and with two datasets the framework is not applicable in its intended form at all.

Table 4.2: Ablation on Bank Marketing: PR-AUC of each pre-processing step applied in isolation to the raw data, and of the complete pipeline. Downstream model: logistic regression.

Stage	Operation	PR-AUC
0	Raw input (label encoding only)	0.4124
1	Imputation only	0.4124
1	One-hot encoding only	0.4098
1	Standardization only	0.4415
1	Winsorization only	0.3884
1	Box-Cox only	0.4394
1	PCA 90 % only	0.4044
1	PCA 20 % only	0.3303
1	Domain features only	0.4121
1	Interaction terms only	0.4494
1	DAE encoding only	0.4223
1	VAE encoding only	0.4035
1	FT-Transformer encoding only	0.4586
2	Pipeline A complete	0.4519
2	Pipeline B complete (PCA 90 %)	0.4185
2	Pipeline B complete (PCA 20 %)	0.2700
2	Pipeline C complete	0.4508
2	Pipeline D complete	0.4208
2	Pipeline E complete	0.4064
2	Pipeline F complete	0.4607

Consequence. That 401 of 595 comparisons on Bank Marketing reach significance while the effect sizes concerned lie between 0.002 and 0.016 PR-AUC at fold standard deviations of 0.009 to 0.020 is the visible symptom of the problem rather than a finding. A further symptom is the discreteness of the statistic: on 15 paired samples the minimum attainable p -value is $2/2^{15} \approx 6.1 \times 10^{-5}$, and a large share of comparisons attains exactly this floor, so the test cannot rank differences by magnitude even where it is applicable. The significance analysis is accordingly not treated as evidence anywhere in this work. Every statement in this chapter rests on the difference in mean PR-AUC read against the fold standard deviation, and differences smaller than one fold standard deviation are reported as undecided regardless of what the test returns.

4.4 Ablation: Which Step Carries the Effect?

The ablation isolates each pre-processing operation by applying it alone to the raw data. Table 4.2 reports the results for Bank Marketing, using logistic regression as the downstream proxy.

The Bank Marketing ablation provides the sharpest result in this work. The corresponding table for Online Retail is given in Appendix A.2.

Fig. 4.10 presents the same decomposition as a matrix, with one column per pipeline, and Fig. 4.11 reduces it to the single quantity that matters for the comparison, the gain of each complete pipeline over the raw input. Fig. 4.11(a) and (b) should be read with their differing horizontal scales in mind: the Bank Marketing panel spans 0.19 PR-AUC, the Online Retail

panel 0.003.

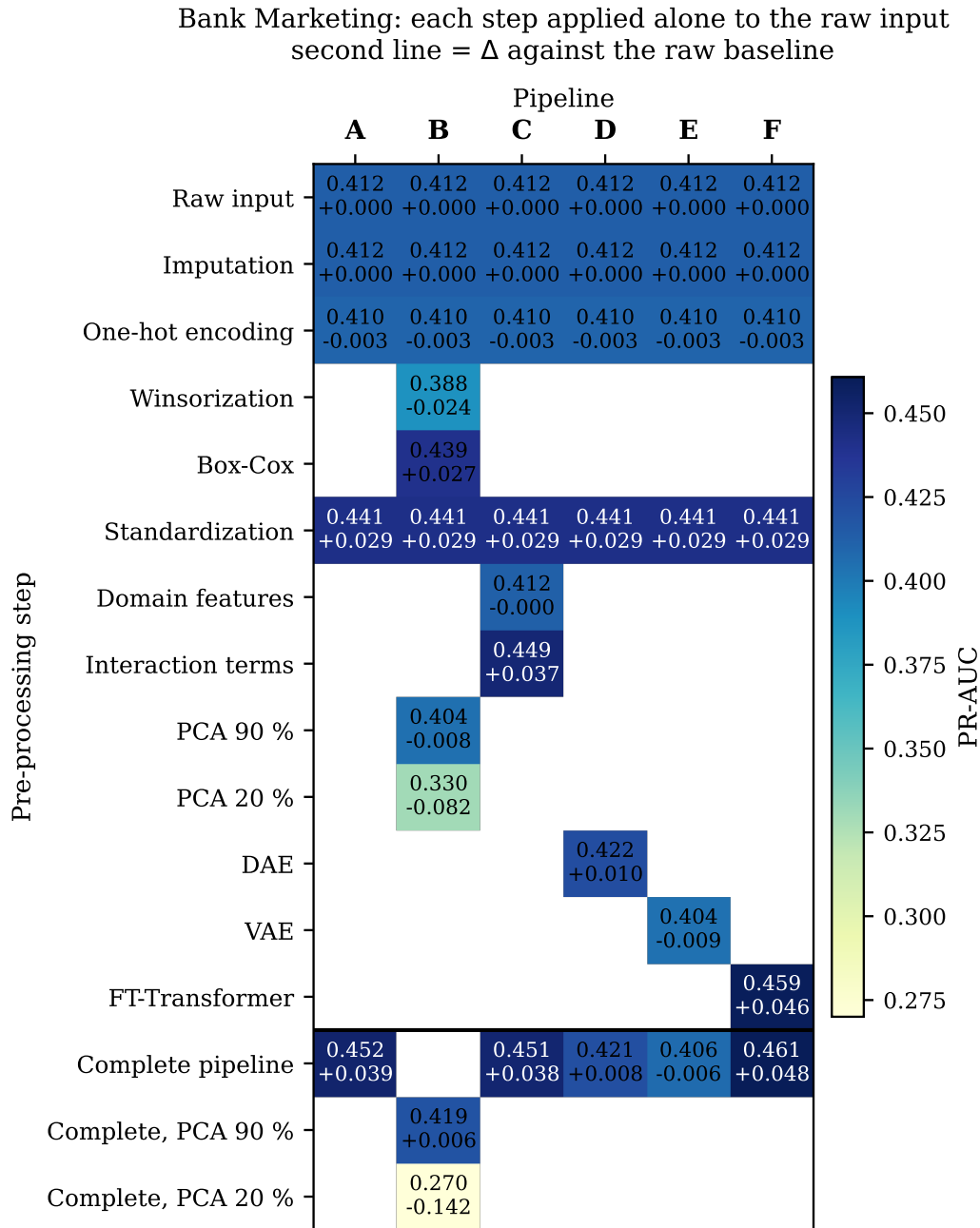


Figure 4.10: PR-AUC of each pre-processing operation applied on its own to the raw input of Bank Marketing, by pipeline, with logistic regression downstream. The upper number in a cell is the PR-AUC and the lower one the difference against the raw baseline of 0.4124. Empty cells mean the operation is not part of that pipeline. The rule separates the isolated steps from the complete pipelines.

Standardization carries most of the achievable benefit. Applied alone it lifts PR-AUC from 0.4124 to 0.4415, a gain of 0.029. The complete Pipeline A, which adds imputation and encoding on top of it, reaches 0.4519, a gain of 0.040. One elementary operation therefore accounts for roughly three quarters of what the assembled baseline achieves. Four of the seven complete pipelines gain less over the raw input than standardization does on its own:

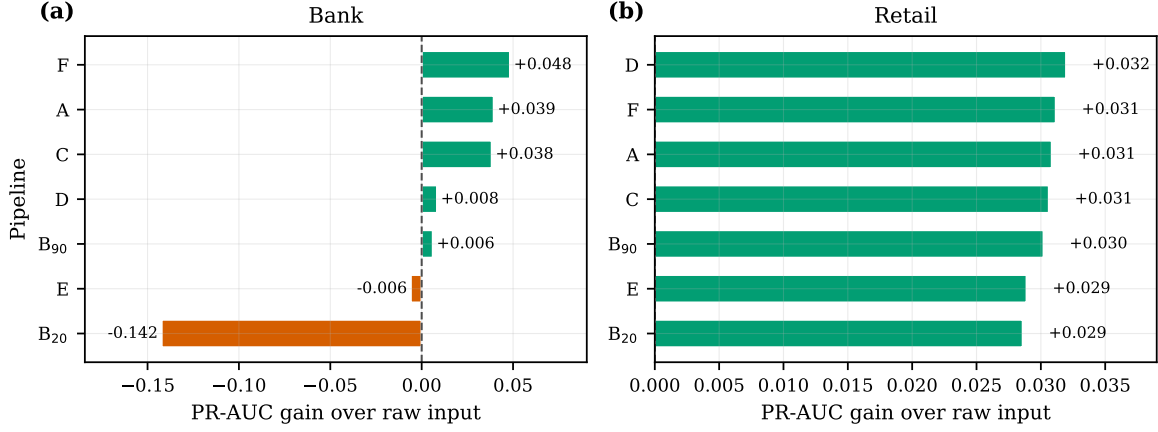


Figure 4.11: Difference in PR-AUC between each complete pipeline and the raw baseline for (a) Bank Marketing and (b) Online Retail, sorted by magnitude. Green marks a gain, orange a loss. Note the differing horizontal scales.

Pipeline B at 90 % variance (0.006), Pipeline D (0.008), Pipeline E (-0.006) and Pipeline B at 20 % variance (-0.142).

Only the most elaborate operation in each family exceeds standardization on its own, interaction terms reaching 0.4494 (0.037 over raw) and the FT-Transformer 0.4586 (0.046). The second figure deserves attention: the complete Pipeline F reaches 0.4607, only 0.002 more than its encoder applied to raw input. Whatever Pipeline F achieves, it achieves through the encoder alone, and the imputation, encoding and scaling that precede it are close to redundant.

Imputation contributes nothing on this dataset. Bank Marketing has no missing values, so this is expected and serves as a sanity check on the ablation implementation.

One-hot encoding alone is mildly harmful. Applied without subsequent standardization, one-hot encoding reduces PR-AUC from 0.4124 to 0.4098. The reason is scale imbalance: binary indicators taking values in $\{0, 1\}$ coexist with raw numeric features spanning thousands of units, and the logistic regression gradient is dominated by the latter. This illustrates that pre-processing steps are not independently beneficial, since their value depends on what follows them.

Winsorization is harmful on Bank Marketing but helpful on Online Retail. Applied alone it costs 0.024 PR-AUC on Bank Marketing (0.3884 versus 0.4124) while gaining 0.036 on Online Retail (0.7706 versus 0.7350). The difference is explained by the distributions: Online Retail’s monetary features are extremely right-skewed with genuine extreme values that dominate the scaling, whereas Bank Marketing’s numeric features contain few outliers but several bounded variables that winsorization merely truncates.

On Online Retail, standardization is again the decisive step. Raw input provides 0.7350 and standardization alone 0.7659, after which almost nothing moves: interaction terms reach 0.7666, PCA at 90 % variance 0.7665, the denoising autoencoder 0.7669 and the complete Pipeline C 0.7656. Only winsorization and Box-Cox, at 0.7706 and 0.7696, add materially,

which is consistent with the extreme right skew of the monetary features. The four RFM aggregates otherwise leave little for further transformation to extract.

Composition is not additive. Pipeline B in full reaches 0.4185, well below its Box-Cox step alone (0.4394) and below its standardization step alone (0.4415), because the PCA that terminates the chain discards directions the earlier steps had prepared. Pipeline C in full reaches 0.4508, barely above its interaction-term step alone (0.4494), so the domain features and the standardization that surround the interaction terms contribute 0.001 between them. Steps interact, and assembling them in sequence does not accumulate their individual gains. The composed Pipeline C in fact falls marginally below the composed Pipeline A here, at 0.4508 against 0.4519, and by 0.0002 on Online Retail. Under the untuned ablation protocol the interaction terms therefore buy nothing once they are embedded in the full chain. The advantage they produce in the main experiment appears only when the downstream model is tuned and can exploit them, which is a further instance of the protocol dependence discussed at the end of this section.

The domain features themselves contribute nothing. This is the most consequential single reading of the ablation and it holds on both datasets. On Bank Marketing the domain features applied alone reach 0.4121 against 0.4124 for the raw input, a difference of -0.0002 and thus a null result. On Online Retail they reproduce the raw value to five decimal places (0.73496 in both cases), because the RFM aggregation that Section 3.3.3 declares as the domain component is not specific to Pipeline C at all: the raw dataset is transactional and carries no customer-level label, so every one of the seven pipelines operates on the same four RFM aggregates. Nothing distinguishes Pipeline C there except the interaction terms.

The entire advantage of Pipeline C therefore rests on the polynomial interaction terms, which reach 0.4494 against 0.4124 for raw input on Bank Marketing. That operation is generic. It forms pairwise products of standardized numeric features and requires no subject-matter knowledge whatsoever.

Online Retail supplies the cleanest form of this argument, because there the two pipelines are separated by nothing else. With the domain step contributing exactly zero, Pipeline C is Pipeline A plus interaction terms and nothing further, and in the main experiment it leads by 0.0090 in the model mean and for every one of the five classifiers individually, from 0.0068 for the random forest to 0.0118 for logistic regression. Whatever consistent advantage Pipeline C holds is thus attributable to the interaction terms alone, with no confounding step left to carry it. Table 4.4 confirms the reading independently: the most important feature of Pipeline C on Bank Marketing is the product `cons.price.idx` $\times$ `nr.employed`, carrying 36.9 % of the total gain, which is an interaction between two macroeconomic indicators rather than a hand-crafted domain feature. The finding of this study is consequently not that domain knowledge pays off, but that *polynomial interaction terms on standardized features* do, and that conclusion is the more useful of the two for practice, since it presupposes no domain expert.

Taken together, the ablation supports a conclusion that the aggregate performance table only hints at: *the elaborate parts of these pipelines are largely not where the performance comes from*. A practitioner who applies imputation, encoding and standardization correctly has captured about three quarters of what any classical pipeline in this study delivers. The one operation that clearly surpasses this is the supervised encoder of Pipeline F.

Table 4.3: Robustness: mean relative change in PR-AUC averaged over all corruption rates per mechanism, with the worst single observation in the final column. Corruption is applied to the training fold only, while the validation fold remains clean. Downstream model: logistic regression.

Pipeline	Base	MCAR	MAR	MNAR	Outlier	Noise	Worst
<i>Bank Marketing</i>							
A	0.452	−2.4	−4.0	−8.2	−8.3	−2.0	−12.5
B (PCA 90 %)	0.419	−1.5	−2.0	−0.4	+1.4	−0.2	−7.0
B (PCA 20 %)	0.270	−4.9	−9.1	+14.7	+9.6	−4.5	−14.0
C	0.451	−2.2	−30.9	−16.9	−7.5	−2.8	−60.1
D	0.427	−0.7	−1.3	−27.7	−14.1	−0.3	−31.5
E	0.405	−2.3	−4.7	−34.0	−4.9	−1.0	−45.0
F	0.476	−2.8	−4.5	−6.6	−1.0	−1.4	−17.0
<i>Online Retail</i>							
A	0.766	−1.4	−1.8	−4.7	−1.5	−0.6	−12.8
B (PCA 90 %)	0.765	−3.0	−5.4	−9.8	±0.0	±0.0	−25.4
B (PCA 20 %)	0.764	−3.4	−5.8	−7.0	−0.7	±0.0	−25.3
C	0.766	−1.4	−2.2	−8.9	−1.3	−0.9	−18.6
D	0.766	−1.4	−3.5	−11.8	−0.7	−0.6	−28.3
E	0.766	−1.4	−1.8	−7.8	−0.9	−0.3	−27.9
F	0.770	−1.6	−3.7	−9.2	−0.5	−1.4	−30.4

All values in percent relative to the uncorrupted baseline.

That exception deserves more than a footnote, because it inverts the headline result. At 0.4586 the supervised encoder is the strongest single operation in the study, ahead of interaction terms at 0.4494 and of standardization at 0.4415, and the complete Pipeline F reaches 0.4607, the highest value in the table. Under the ablation protocol, then, the learned representation wins. Under the tuned protocol of Section 4.2 it loses by 0.016.

Both protocols are defensible, and they answer different questions. The ablation holds the downstream model fixed and untuned, which isolates the contribution of the representation itself and gives no pipeline a tuning advantage. The main experiment tunes each downstream model, which is what a practitioner would do and which rewards representations that a strong learner can exploit. The classical pipelines gain more from that tuning than the encoders do, and the ordering reverses. The disagreement between the two is itself a result: the deficit of the learned representations is not a property of the representations alone but of the combination of representation and downstream budget, and a study reporting only one protocol would have reached a confident but incomplete conclusion.

4.5 Robustness to Data Quality Problems

Table 4.3 reports the relative change in PR-AUC under each corruption mechanism on Bank Marketing.

The base column of Pipelines D–F is not comparable with that of A–C. This has to be stated before the table is read, because the discrepancy is visible in it. Pipeline F enters the robustness sweep at a base value of 0.476, above both the main experiment (0.4572) and the ablation (0.4607). The reason is that the robustness harness loads the encoder weights

serialized by the standalone pipeline scripts, and those encoders are fitted once on the entire training split. They are therefore the single-pass encoders whose evaluation Section 3.6.3 identifies as leaking, not the per-fold encoders of the corrected main experiment. Every cross-validation fold of the sweep is encoded by a model that has already seen that fold, and for the supervised Pipeline F including its labels. The inflated base value is the direct signature of this.

The consequence is drawn explicitly. Re-running the sweep with per-fold encoders was not within the compute budget of this work, so the affected comparisons are withdrawn rather than repaired. No statement in this study compares the robustness of Pipelines D–F against that of Pipelines A–C, and the placement of the AI pipelines in a common robustness ranking is not claimed. What remains valid is the comparison *among* D, E and F, since all three are treated identically by the harness and the same distortion applies to each, and the comparison among A, B and C, which involves no encoder and is unaffected. RQ2 is answered on that restricted basis in Section 4.9.2.

A note on how these values are aggregated. Degradation is reported as the mean over all corruption rates rather than at a single rate, because several curves are strongly non-monotonic, as Fig. 4.14 shows. On Bank Marketing under MNAR, for instance, Pipeline E falls from 0.405 to 0.223 at a corruption rate of 5 %, recovers to 0.321 at 30 %, and falls again to 0.233 at 50 %. Reading such a curve at one point produces arbitrary conclusions, because at 30 % Pipeline F appears to lose only 4.2 % under MNAR, whereas its worst observation is a loss of 17.0 % at 50 %.

The non-monotonicity has a plausible cause. At low rates the MNAR mechanism masks only the extreme upper tail of a feature and replaces it with the median, which distorts the very region where the discriminative signal is concentrated. At higher rates a large share of the feature becomes constant, the downstream model recognizes it as uninformative and down-weights it in favour of the remaining features. A partially corrupted feature can therefore be more harmful than an obviously useless one. At 50 % the additional effect appears that columns cross the 50 % drop threshold and are removed entirely, which changes the mechanism qualitatively.

Five patterns emerge from the aggregated view.

MNAR is the most damaging mechanism for most configurations. For five of the seven pipelines on Bank Marketing and for every pipeline on Online Retail, missingness that depends on the unobserved value causes substantially larger degradation than MCAR or MAR. This is the expected ordering [22]: median imputation is unbiased under MCAR, approximately correctable under MAR given the observed covariates, and systematically biased under MNAR, because the imputed value is drawn from a distribution that excludes the very region where the missing values lie. MCAR, by contrast, is nearly harmless throughout, since only the B_{PCA20} Pipeline loses more than 3.4 % on average under it. Label noise is equally mild, consistent with the downstream models being regularized or ensemble methods with some tolerance for mislabelled instances.

The exceptions are worth naming, because the largest one is not an anomaly but a consequence of the mechanism described below. Pipeline C loses 30.9 % under MAR against 16.9 % under MNAR, so for that pipeline the milder mechanism is by far the more harmful. The reason

is that MAR removes values conditional on other observed columns, which strikes precisely the base features from which Pipeline C derives its ratios and interaction terms, and every derived feature inherits the damage. PCA at 90 % retained variance shows the same reversal on a much smaller scale, at 2.0 % under MAR against 0.4 % under MNAR. B_{PCA20} appears to improve under MNAR, which is the artifact discussed at the end of this section.

PCA-based compression is the most stable on Bank Marketing, at a permanent cost. B_{PCA90} loses 0.5 % on average across all mechanisms and is marginally *improved* by outlier injection. PCA projects the input onto its dominant variance directions, and injected noise largely occupies the directions that are discarded. This stability, however, is purchased by permanently sacrificing 0.044 PR-AUC on clean data. The comparison between B and C is therefore a genuine trade-off rather than a ranking.

Notably this advantage does not transfer to Online Retail, where B_{PCA90} is among the *least* robust configurations under MNAR (−9.8 % on average, worst case −25.4 %). With only four input features there are no redundant variance directions to discard, so the mechanism that provides the robustness on Bank Marketing is unavailable.

The autoencoders are the most fragile of the learned representations under systematic distortion. Pipelines D and E lose 27.7 % and 34.0 % on average under MNAR on Bank Marketing, with worst cases of 31.5 % and 45.0 %. Both are nevertheless almost unaffected by MCAR (−0.7 % and −2.3 %) and by label noise. How these magnitudes compare with those of the classical pipelines is not stated, for the reason given at the beginning of this section.

The contrast between the mechanisms is instructive. Both encoders are trained to reconstruct a *specific* input distribution, and the resulting latent space is only meaningful for inputs drawn from it. Random noise spread evenly across features stays close enough to that distribution, while the systematic distortions of MNAR, in which entire regions of the feature range are replaced by a single value, push inputs off the learned manifold, and the encoder maps them to latent positions that no longer correspond to their class membership.

Pipeline C is the least robust classical configuration on Bank Marketing. Its mean degradation of 12.0 % across all mechanisms is the worst of the classical pipelines, against 5.0 % for the baseline and 0.5 % for B_{PCA90} , driven by MAR (−30.9 % on average) and MNAR (−16.9 %), with a worst single observation of −60.1 %. Its interaction terms and derived ratios are computed from corrupted inputs, so a single degraded base feature propagates into every feature derived from it. Pipeline C thus combines the best clean-data performance with the greatest sensitivity to input quality, the sharpest trade-off in the study.

The effect is confined to the feature-rich dataset. On Online Retail, Pipeline C degrades by 3.0 % and is the second most robust classical configuration there, behind the baseline at 2.0 % and ahead of both PCA variants. Four RFM aggregates leave little room for derived features to amplify corruption, so the fragility is a property of elaborate feature construction on rich input rather than of the pipeline as such. The claim is accordingly stated for Bank Marketing and not generalized.

The FT-Transformer is the most robust of the learned representations. On Bank Marketing it degrades by 3.3 % on average across all mechanisms, against 8.8 % for the de-

noising autoencoder and 9.4% for the variational one, and its advantage under MNAR is substantial and consistent at 6.6% against 27.7% and 34.0%. Within the family of learned representations, then, the training objective matters more for robustness than the architecture: F’s supervised signal optimizes for class discrimination rather than faithful reconstruction, and its token-wise attention permits a degraded feature to be down-weighted relative to the others. Whether this places F above or below any classical pipeline is a question the harness cannot answer, since all three encoders enter the sweep with an inflated baseline while the classical pipelines do not.

This advantage should not be overstated. F’s worst single observation on Bank Marketing is a loss of 17.0%, and on Online Retail it is the largest of all seven configurations (−30.4%). Averaged over mechanisms, F ranks fourth of seven on Online Retail. The claim that survives both datasets is the narrower one: *among learned representations*, the supervised encoder is clearly the most robust, while against the classical pipelines the picture is dataset-dependent.

An artifact to note. B_{PCA20} shows apparent *improvements* under MNAR (+14.7% on average) and outlier injection (+9.6%) on Bank Marketing. This is not a robustness finding. Its uncorrupted baseline of 0.270 is so degraded by compression to four components that injected corruption happens to shift the retained variance directions in a marginally more informative way. Relative changes computed against a near-degenerate baseline are not interpretable, and this row should be disregarded.

Bank Marketing: PR-AUC under increasing corruption

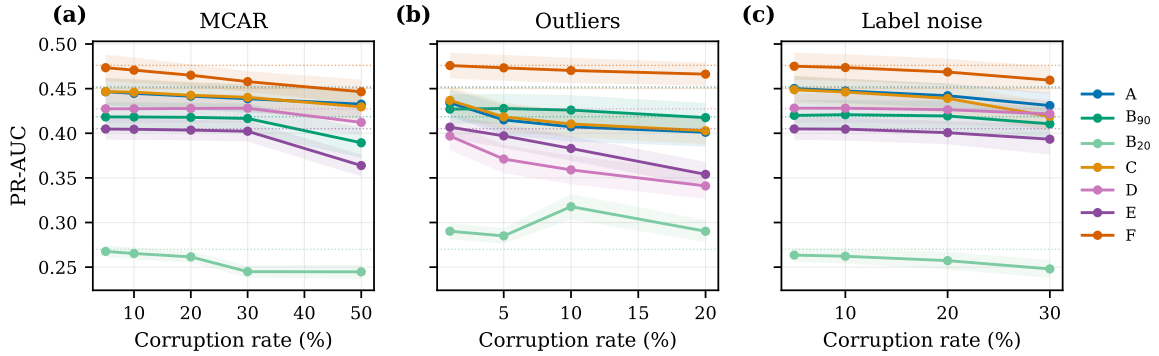


Figure 4.12: PR-AUC on Bank Marketing under (a) missingness completely at random, (b) injected outliers and (c) label noise. Each curve is one pipeline, the band spans one standard deviation over the 15 folds and the dotted lines mark the uncorrupted baseline of the corresponding pipeline. The absolute height of a curve reflects the uncorrupted performance of that pipeline, not its robustness, which is the relative distance to its own dotted baseline.

Fig. 4.12 shows that the milder mechanisms are not uniformly mild. Panel (a) keeps all seven curves flat until a rate of 30% and then produces a step at 50%, the point at which columns cross the drop threshold and are removed altogether. Panel (b) separates the field, with F and B_{PCA90} nearly horizontal while D and E fall by roughly a seventh between the mildest and the strongest rate tested, so outlier injection discriminates between pipelines in a way that random missingness does not.

The contrast between Fig. 4.12(a) and Fig. 4.13(a) is the clearest dataset effect of this section.

Online Retail: PR-AUC under increasing corruption

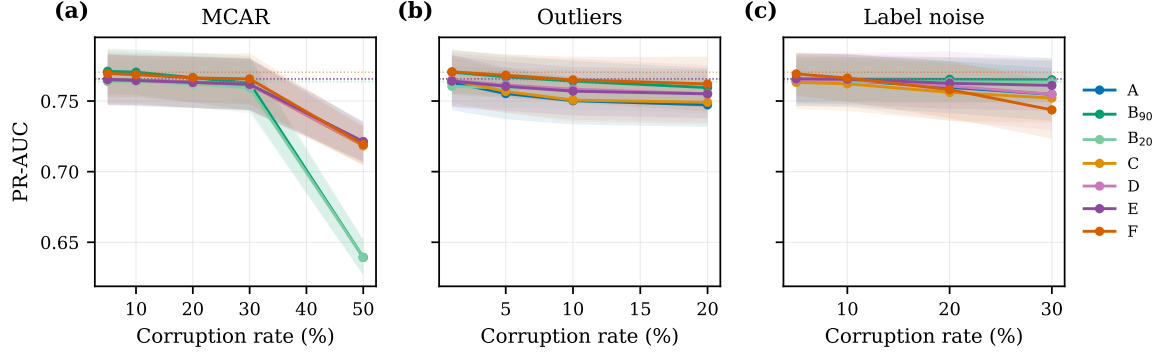


Figure 4.13: PR-AUC on Online Retail under (a) missingness completely at random, (b) injected outliers and (c) label noise, drawn as in Fig. 4.12. As there, the absolute height of a curve is the uncorrupted performance of that pipeline and not its robustness.

On Bank Marketing the PCA variants absorb missingness without visible loss, while on Online Retail they fall to 0.639 at a rate of 50 % against roughly 0.72 for every other pipeline, because four input features leave no redundant directions into which the injected noise could be projected away. Elsewhere Fig. 4.13 is unremarkable, and that is itself the finding, since the seven curves stay within 0.02 of one another across both remaining panels and reproduce under corruption the compression already seen on clean data.

Bank Marketing: MNAR missingness

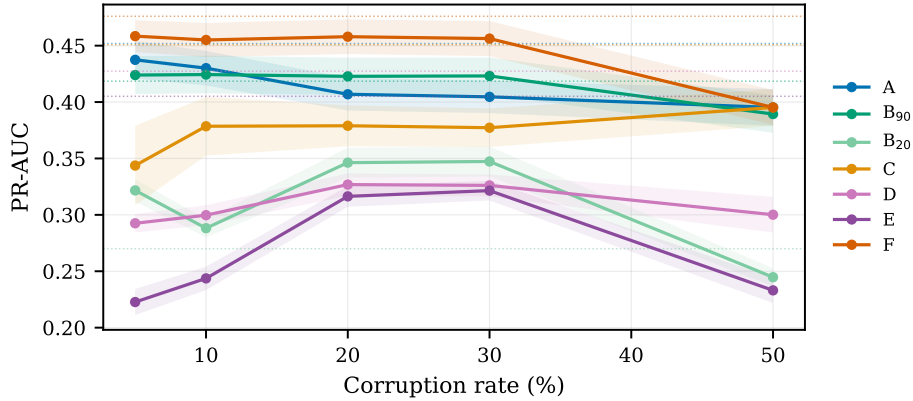


Figure 4.14: PR-AUC on Bank Marketing under missingness not at random, at rates from 5 to 50 %. Bands span one standard deviation over the 15 folds and the dotted lines mark the uncorrupted baselines.

Fig. 4.14 makes the non-monotonicity concrete. Pipelines C, D and E reach their *lowest* PR-AUC at the mildest corruption rate tested and partially recover as the rate rises, while A, B_{PCA90} and F decline conventionally and reach their minimum at 50 %. The three pipelines that break the expected pattern are exactly those whose features are derived or learned rather than passed through, which supports the manifold argument given above.

4.6 Business Impact

PR-AUC quantifies model quality but does not communicate decision value. The lift at the top decile translates it: it states how many times more responders a campaign reaches by contacting the highest-scoring 10 % of the population instead of a random 10 %.

Fig. 4.15 shows the full curves. They separate only in the leftmost region, which is where a campaign operates, and converge once more than half of the base has been contacted.

On Bank Marketing the best pipelines achieve a lift of approximately 4.5. A campaign contacting the top decile therefore reaches four and a half times as many subscribers as an untargeted campaign of the same size, an operationally decisive improvement. The differences between pipelines translate directly: Pipeline C reaches 4.55, B_{PCA20} only 3.50, a difference of roughly one fifth of the campaign’s yield.

On Online Retail the lift is approximately 1.85 for *every* pipeline. The retention target is far less concentrable, because customers who will return are spread much more evenly across the RFM feature space than term-deposit subscribers are across the bank’s client attributes. This is a property of the problem, not a deficiency of the models, and it carries a practical message, namely that for this kind of churn problem the choice of pre-processing has almost no effect on campaign economics, so effort is better invested elsewhere.

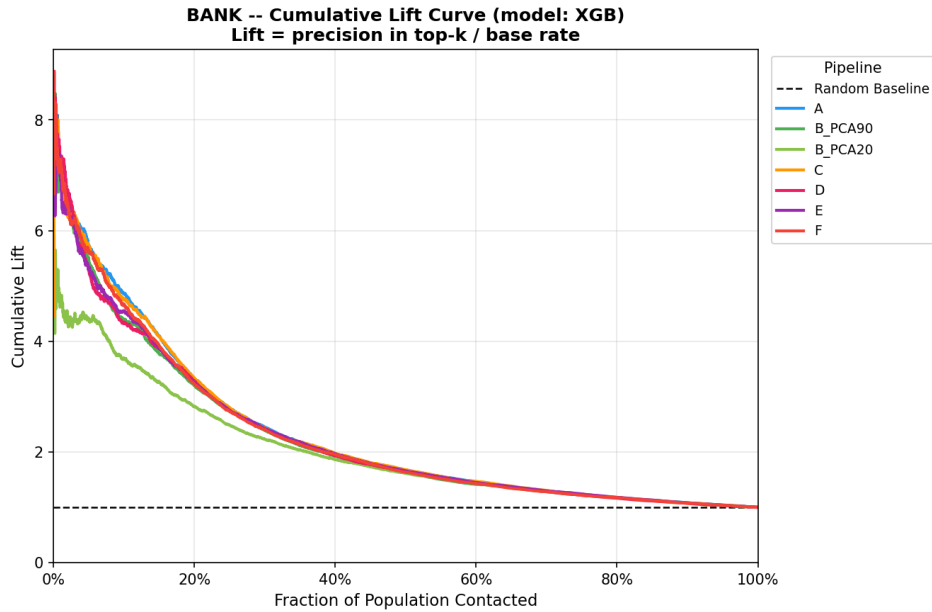


Figure 4.15: Cumulative lift on Bank Marketing with XGBoost downstream. Customers are ranked by predicted probability and the horizontal axis gives the fraction contacted. The dashed line at one corresponds to random targeting.

4.7 Interpretability of the Representations

A dimension that does not appear in any performance metric is whether the resulting model can be explained to the business unit that has to act on it. Table 4.4 lists the most important feature of each pipeline, measured by XGBoost gain on Bank Marketing.

Table 4.4: Most important feature per pipeline (XGBoost gain, Bank Marketing).

Pipeline	Top feature	Share of gain
A	<code>nr.employed</code>	38.3 %
C	<code>cons.price.idx × nr.employed</code>	36.9 %
B (PCA 90 %)	PC1	14.7 %
D	<code>dae_4</code>	19.7 %
E	<code>vae_13</code>	22.0 %
F	<code>ftt_42</code>	21.8 %

The three families differ qualitatively. Pipelines A and C provide named, actionable drivers, since employment level and consumer price index are quantities a marketing department understands, can monitor, and can incorporate into campaign timing. Pipeline B provides principal components which are opaque at first glance but remain inspectable, since the loadings can be decomposed back onto the original variables. Pipelines D, E and F provide axes such as `dae_4` that carry no semantic content whatsoever.

A second pattern is visible in the concentration of importance. The classical pipelines concentrate 38 % of total gain in a single feature, the encoders at most 22 %, distributing information across their latent dimensions. This is expected behavior for a learned compression, but it compounds the interpretability problem, because not only is each axis meaningless, no small subset of axes dominates.

The practical consequence is that the AI pipelines in this study offer no performance advantage while removing the ability to explain the model. In a Business Intelligence context, where model outputs inform human decisions and frequently require justification to stakeholders or regulators, this asymmetry is decisive.

Fig. 4.16 makes the interpretability gap concrete. Panel (a) and (b) name quantities a marketing department can act on, (c) yields principal components that can at least be traced back to their loadings, and (d) to (f) yield axes carrying no business meaning at all.

4.8 Scaling with Training Set Size

A recurring argument in favour of representation learning is that its advantages emerge with scale. If the deficit observed here were a consequence of insufficient training data, the performance gap should narrow as the training set grows.

Learning curves were computed for the classical pipelines by evaluating each at training fractions from 10 % to 100 %. On Bank Marketing, Pipeline A rises from 0.424 at 10 % of the data to 0.455 at 90 % and is essentially flat beyond 50 %. Pipeline C rises from 0.413 to 0.460 and is still improving slightly at the largest fraction, consistent with its higher feature count requiring more data to exploit. On Online Retail both curves are flat throughout, which is expected given four predictors.

Fig. 4.17 shows both datasets. The flatness of Fig. 4.17(b) beyond the smallest fraction is expected with four predictors, while Fig. 4.17(a) still separates the pipelines at every training size.

Top features per pipeline, Bank Marketing

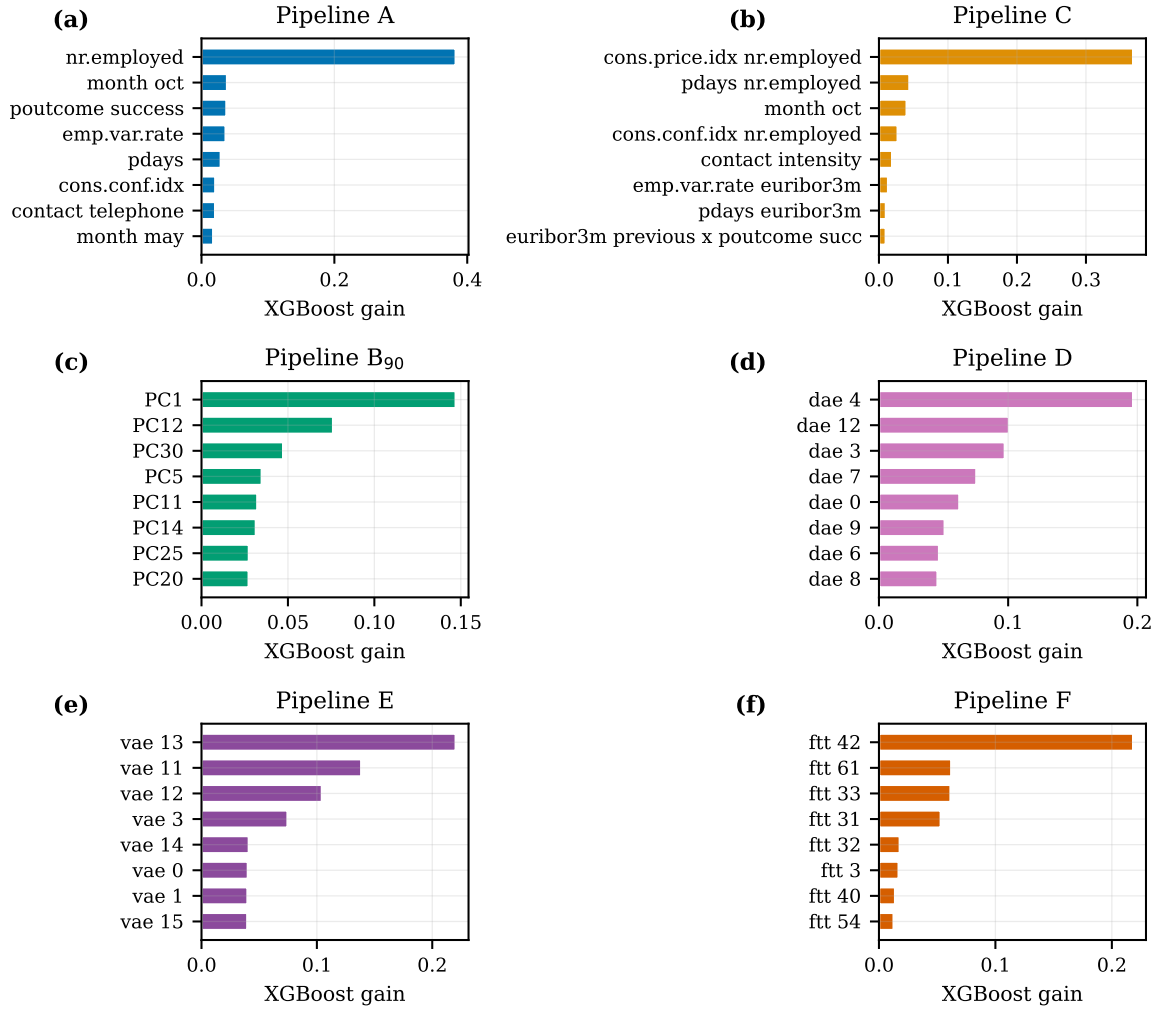


Figure 4.16: The eight highest ranked features by XGBoost gain on Bank Marketing for Pipelines (a) A, (b) C, (c) B_{PCA90}, (d) D, (e) E and (f) F. Bar lengths are comparable within a panel but not between panels.

The saturation is informative in itself: at the sizes available here, additional data would not change the ranking of the classical pipelines. Whether it would change the position of the learned representations cannot be answered from these data, for the reason given below.

A restriction on the AI pipelines. The learning curves for Pipelines D–F are computed from the exported encoder representations and are therefore subject to the same leakage mechanism described in Section 3.6.3: the encoder was fitted on the full training split, so every evaluation point re-uses data the encoder has already seen. The resulting values are inflated (Pipeline F peaks at 0.492 on Bank Marketing against a leakage-free 0.457) and are consequently not reported here. Producing valid learning curves for the AI pipelines would require re-fitting the encoder at each training fraction, which was outside the available compute budget. This is recorded as a limitation in Section 4.10.

PR-AUC against training set size, classical pipelines only

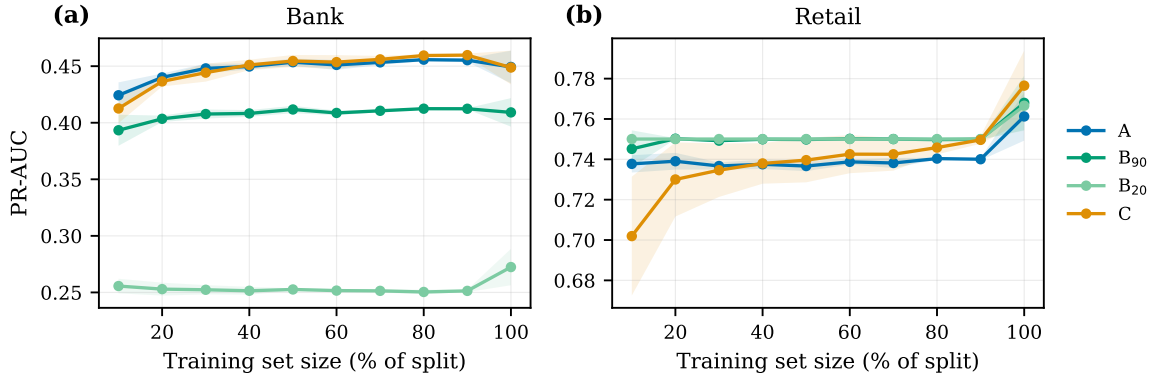


Figure 4.17: PR-AUC against the fraction of the training split used, at ten points from 10 to 100 %, for (a) Bank Marketing and (b) Online Retail. Bands span one standard deviation over folds. Only the classical pipelines are shown.

4.9 Discussion

The preceding sections reported what was measured. This section interprets those measurements against the two research questions and against the existing literature, and adds two findings that concern the methodology rather than the substantive question.

4.9.1 RQ1: Predictive Performance

AI-based representation learning does not outperform classical pre-processing on either dataset. Pipeline C exhibits the highest PR-AUC on both (0.4728 and 0.7797), the plain baseline is a close second, and the best AI pipeline trails the leader by 0.016 on both datasets, or the baseline by 0.013 on Bank Marketing and 0.008 on Online Retail.

What produces the lead of Pipeline C is not what its name suggests. The ablation attributes the whole of it to the polynomial interaction terms and none of it to the domain features, which reach 0.4121 against 0.4124 for raw input on Bank Marketing and reproduce the raw value exactly on Online Retail. The defensible formulation of this result is therefore that interaction terms on standardized features win, not that domain knowledge wins. Since interaction-term construction is a generic operation available without a subject-matter expert, the practical reach of the finding is wider than the pipeline label implies, and the recommendation in Section 4.11 is conditioned accordingly.

How firmly the leader is established differs between the datasets. On Bank Marketing Pipeline C is ahead of the baseline by 0.0025, which is a fifth of the smallest fold standard deviation in the comparison, and it wins for only two of the five downstream classifiers. The two pipelines are not distinguishable there, the comparison between them has to be recorded as undecided, and the ordering in Table 4.1 must not be read as a result on that dataset. On Online Retail the lead is 0.0090 in the mean over the five classifiers and holds for every one of them, so the direction is consistent even though each individual difference remains inside the fold spread. What the study does establish firmly is the negative result: no learned representation reaches the classical family, at a margin of 0.016 that is identical on both datasets and

sits at the upper edge of the fold spread.

This is consistent with the body of evidence on tabular data. Grinsztajn et al. [7] and Shwartz-Ziv and Armon [8] report that tree-based classifiers retain an advantage over deep architectures on tabular problems, and Borisov et al. [45] reach a comparable conclusion in their survey. Those studies vary the classifier while holding the input fixed. The contribution of this work is to show that the same conclusion holds along the other axis, with the deep architecture used purely as a *pre-processing operator* rather than as an end-to-end classifier. The classifier-level finding is itself visible here only on Bank Marketing, where a tree ensemble is the best downstream model for every one of the seven pipelines, whereas on Online Retail logistic regression or the perceptron enters the leading three for all of them. The representations it learns do not carry information that classical transformations fail to expose.

The gap plausibly has two causes. The encoders compress into 16 or 64 dimensions an input that already possesses a well-structured representation after standardization, so the bottleneck can only discard information, and the reconstruction objective of D and E is not aligned with class discrimination [9]. Beyond that the training signal is limited, since Online Retail provides only 3,616 samples with four features, far below the regime in which representation learning typically demonstrates its advantages.

Pipeline F’s position supports this reading. As the only encoder with a supervised training signal, it outperforms both autoencoders on Bank Marketing (0.457 against 0.431 and 0.434). On Online Retail the advantage is only partial: F stays ahead of E but 0.0006 behind D, a difference far inside the fold standard deviation. Alignment between the encoder objective and the downstream task matters more than architectural sophistication.

Pipeline F is not on equal footing with the others, and this cuts both ways. Sections 2.7.3 and 3.6.3 treat the supervised encoder as a leakage hazard and eliminate it by re-fitting per fold. That addresses validity, but it does not remove an asymmetry of the comparison itself: even implemented without leakage, F sees the training labels of its fold while building the representation, whereas A to E see only the features. It is given a training signal that the design withholds from the other six pipelines, which is a real advantage and not merely a risk. Two consequences follow. The failure of F to win under the tuned protocol is a stronger negative result than it appears, since the pipeline loses by 0.016 despite the extra information. Conversely its victory in the ablation, where at 0.4586 it is the strongest single operation of the study, is partly attributable to that same advantage and cannot be credited solely to the tuning protocol as Section 4.4 presents it. A comparison that equalised the information available to every encoder would have to withhold the labels from F as well, at which point it would no longer be the architecture this study set out to evaluate.

4.9.2 RQ2: Robustness

Robustness is the dimension on which the ordering established by RQ1 does not carry over, but the result is more dataset-dependent than the clean-data comparison and does not support a single ranking. It is also the dimension on which this study answers a narrower question than it set out to, for the reason given at the start of Section 4.5: the robustness harness encodes with the single-pass encoders, so Pipelines D–F cannot be placed on a common scale with A–C. RQ2 is therefore answered within the classical family and within the learned family, and not across the two.

Among the classical pipelines the ordering of corruption mechanisms follows theory [22] with one systematic exception. MNAR is the most damaging mechanism, because median imputation is unbiased under MCAR, approximately correctable under MAR, and systematically biased when missingness depends on the unobserved value, while MCAR and label noise are nearly harmless at the rates tested. Pipeline C is the exception, since MAR damages it roughly twice as much as MNAR: MAR removes values conditional on other observed columns, which strikes exactly the base features from which its derived features are computed.

The sharpest trade-off in the study follows from that mechanism. On Bank Marketing, Pipeline C, the strongest performer on clean data, is the least robust of the classical pipelines by a wide margin, at a mean degradation of 12.0 % against 5.0 % for the baseline and 0.5 % for B_{PCA90} . The derived features that produce its clean-data advantage also propagate corruption from their inputs, so the same property accounts for both. On Online Retail the trade-off does not appear: Pipeline C degrades by 3.0 % and is the second most robust classical configuration there, behind the baseline at 2.0 % and ahead of both PCA variants. The fragility is thus a property of elaborate derived features on rich input, not of the pipeline as such, and it argues against selecting a pipeline on benchmark performance alone wherever such features are built.

Within the family of learned representations, where the shared protocol makes the comparison valid, the training objective governs robustness far more than the architecture. The supervised FT-Transformer degrades by 6.6 % on average under MNAR against 27.7 % and 34.0 % for the two autoencoders. Optimizing a representation for class discrimination rather than for faithful reconstruction makes it substantially less dependent on inputs matching the training distribution, while the autoencoders' latent spaces are only meaningful for inputs drawn from the distribution they were fitted on.

Beyond these the picture diverges by dataset. On Bank Marketing, PCA-based compression is the most stable classical configuration, discarding the variance directions in which injected noise predominantly lies, a passive robustness bought by permanently sacrificing 0.044 PR-AUC. On Online Retail the same mechanism is unavailable, because four input features leave no redundant directions to discard, and B_{PCA90} becomes the least robust of the classical pipelines instead.

A methodological point qualifies all of these statements. Several degradation curves are non-monotonic, in some cases severely: Pipeline E under MNAR on Bank Marketing reaches its minimum at a corruption rate of 5 % and partially recovers thereafter. Robustness assessments that evaluate a single corruption rate, which is the common practice, can therefore produce arbitrary orderings. The values reported here are averaged over all rates for this reason, and the worst single observation is stated alongside.

For practice the defensible recommendation is narrower than a ranking. Where input quality is controlled and monitored, Pipeline C remains preferable on performance grounds, but its fragility on feature-rich input makes validation of the base features a prerequisite rather than an optional safeguard. Where data quality cannot be controlled, a compression-based pipeline is the safer choice on feature-rich input. Among the learned representations the supervised encoder is the only one worth considering, since the unsupervised autoencoders offer neither accuracy, nor interpretability, nor robustness. Whether any of them would outperform a classical pipeline under corruption is a question this study cannot answer.

4.9.3 Methodological Findings

Beyond the substantive question, this study produced two methodological findings that are consequential enough to report as results in their own right.

Evaluation design determines the conclusion. The naive evaluation protocol for Pipeline F, which fits the encoder once and cross-validates on its output, provides a PR-AUC of 0.858 on Bank Marketing, against a true value of 0.457. Had this figure been reported, the study would have concluded that representation learning outperforms classical pre-processing by a wide margin. The error is not obvious from the results themselves, since the inflated value is not implausible, the fold variance is small, and the pipeline behaves normally in every other respect. It becomes visible only through the disagreement between the cross-validation estimate and the held-out test split, and is confirmed by re-fitting the encoder inside each fold. Studies that use supervised encoders as feature extractors are exposed to this artifact whenever the extracted features are re-used for cross-validation.

Pre-processing steps interact, and silent failures are plausible. Two further defects were found during development that produced valid-looking but incorrect results: winsorization applied indiscriminately to one-hot dummy variables silently reduced 46 of 62 Bank Marketing features to constants, and an exception handler in the robustness harness substituted a generic transformation whenever the saved encoder failed to load, causing three different pipelines to report identical values. Neither produced an error message. The agreement between the ablation study and the main experiment (0.461 versus 0.457 for Pipeline F on Bank Marketing after correction, against a discrepancy of 0.48 before) served as the diagnostic that confirmed the corrections.

4.10 Limitations

Ten restrictions bound the conclusions drawn above. The first two concern the validity of the evaluation apparatus itself and are the most consequential. The remaining eight concern the scope of the experiment and the compute budget available for it.

The significance analysis is not usable. As set out in Section 4.3, the Wilcoxon signed-rank test was applied to the 15 scores of a repeated cross-validation, which are not independent, so its p -values understate the variance and reject too readily. The corrected resampled t -test [42] would have been the appropriate choice. The test additionally saturates at its discrete minimum of 6.1×10^{-5} for a large share of comparisons and can therefore not rank differences by magnitude in any case. No conclusion in this work rests on the test. All comparisons are decided on differences in mean PR-AUC read against the fold standard deviation, and differences below one fold standard deviation are recorded as undecided.

The robustness results for Pipelines D–F rest on leaking encoders. The harness loads the encoder weights serialized by the standalone pipeline scripts, and those are fitted once on the whole training split rather than per fold, so each validation fold of the sweep was already seen by the encoder that transforms it. This is the same defect that Section 3.6.3 removes from the main experiment, and its signature is the base value of 0.476 for Pipeline F in Table 4.3, which exceeds both the main experiment (0.4572) and the ablation (0.4607). Re-running 24 corruption settings across 15 folds, three pipelines and two datasets with per-fold encoder training was outside the compute budget, so the affected claims are withdrawn

rather than repaired: this work makes no statement about the robustness of the AI pipelines relative to the classical ones. The comparison among D, E and F remains valid, since all three are distorted identically, and the comparison among A, B and C involves no encoder at all.

Two datasets limit generalization. The consistency of the findings across two structurally different datasets is reassuring but does not establish generality. In particular, both datasets are of moderate size, and representation learning may behave differently at considerably larger scale.

The ablation uses a single downstream proxy. Logistic regression was chosen for determinism and speed. Step contributions may differ for tree-based learners, which are for instance insensitive to feature scaling, the operation identified as most valuable in the ablation.

Robustness measures inference-time behavior. Independently of the point above, the encoders are applied frozen to corrupted data, which corresponds to a deployed model encountering degraded inputs. It does not measure how robustly the encoders would *train* on corrupted data, and that question remains open here.

Compute budgets constrain the AI pipelines. Per-fold encoder training is capped at 60 epochs against 100 for single-pass runs, and hyper-parameter optimization uses 20 trials. A larger budget might narrow the gap, although the ablation, in which the encoders are evaluated without any downstream tuning budget, points in the same direction.

Hyper-parameter optimization is not nested. As described in Section 3.5.2, this introduces a mild optimistic bias in absolute cross-validation estimates that applies uniformly across combinations and therefore does not affect the comparison.

Learning curves are unavailable for the AI pipelines. As set out in Section 4.8, valid learning curves for Pipelines D–F would require re-fitting the encoder at each training fraction. The question whether the performance deficit of the learned representations narrows with more data therefore remains open within this study.

Class weighting is not applied uniformly across downstream models. The multi-layer perceptron weights its binary cross-entropy by the inverse class ratio, while logistic regression, random forest, XGBoost and LightGBM are trained without any class weighting, although all four support it. The consequence is visible on Bank Marketing, where the perceptron reaches an F_1 of 0.454 against 0.337 to 0.358 for the other four models and at the same time a Brier score of 0.152 against 0.077: the weighting shifts its decision threshold towards the minority class and sacrifices calibration in doing so. PR-AUC, the primary metric, is threshold-independent and therefore unaffected, and because the same model is applied to every pipeline the comparison between pipelines is not distorted. Comparisons of F_1 or Brier score *across models* should nevertheless be read with this asymmetry in mind. Applying class weighting consistently would have required a complete re-run of all 70 configurations.

Encoder hyper-parameters were not tuned. The architectures of Pipelines D–F use configurations fixed a priori (Appendix A.1) while the downstream models received a tuning budget. This asymmetry is unfavorable to the AI pipelines. It was accepted because tuning three encoder architectures in addition to five downstream models across seven pipelines and two datasets was not feasible, and because the reported deficits (0.013–0.016 PR-AUC against

the best classical pipeline) are of a magnitude that encoder tuning would plausibly narrow but not obviously reverse. The point should nevertheless be weighed when interpreting the RQ1 conclusion.

4.11 Practical Recommendation

Table 4.5 condenses the findings into a selection guide. It is formulated in terms of data characteristics and organisational constraints rather than as a ranking, because the results do not support a single ordering across all criteria.

Its scope has to be stated before it is read. The guide rests on two datasets, and the evidence they contribute is unequal. On Online Retail all seven pipelines fall within 0.017 PR-AUC of one another, so that dataset separates almost nothing and functions mainly as a negative control demonstrating that the differences vanish once the input is reduced to four aggregated features. The discriminating evidence comes essentially from Bank Marketing alone. The table should accordingly be read as a guide for tabular binary classification on data of the size and structure examined here, that is a few tens of thousands of rows with mixed numeric and categorical features, and not as a general prescription. In particular, the recommendation against learned representations is not transferable to high-dimensional, weakly structured, or substantially larger problems, where none of the conditions under which they were tested here apply.

Table 4.5: Pipeline selection guide derived from the results of this study.

Situation	Recommendation	Reasoning
Default starting point, tabular problems of the kind examined here	A (baseline)	Standardization alone accounts for three quarters of what the complete baseline achieves (+0.029 against +0.040 PR-AUC in the ablation). Errors at this level cost more than the absence of any advanced technique.
Numeric features that plausibly interact, input quality controlled and monitored	C, for the interaction terms	Highest PR-AUC on both datasets, with fully interpretable features. The ablation attributes the entire gain to the polynomial interaction terms and none of it to the domain features, so no subject-matter expert is needed. Requires input validation, since it is the most corruption-sensitive pipeline.
Input quality unreliable, feature-rich input	B (90 % variance)	Most stable configuration under all corruption mechanisms on the feature-rich dataset, at a permanent loss of 0.044 PR-AUC. Ineffective when few features are available.
Learned representation required (e.g. for integration into a deep pipeline)	F (FT-Transformer)	The only learned representation with near-competitive accuracy and low corruption sensitivity. Forfeits interpretability entirely.
Any situation	<i>not</i> D or E	Unsupervised autoencoders offered neither accuracy, nor interpretability, nor robustness in this study.
Aggressive dimensionality reduction (20 % variance)	avoid on rich input	Loses 0.14 PR-AUC on 62 features, negligible on 4. The loss scales with the information available to discard.

Chapter 5

Summary and Outlook

This chapter draws the results together, answers the two research questions in turn, and derives a recommendation for practice. It closes with the limitations that bound those conclusions and with three directions in which the study could be extended.

5.1 Summary of Findings

This work compared classical statistical pre-processing against AI-based representation learning for Business Intelligence applications. Six pipelines, one of them instantiated in two variants, were evaluated as pre-processing operators, each feeding five downstream classifiers on two public business datasets, giving 70 combinations assessed over 15 cross-validation splits with PR-AUC as the primary metric.

The central result is that representation learning did not deliver on its promise in this setting. Under the best-of-five reading no learned representation surpassed even the plain baseline of imputation, encoding and scaling, the best of them trailing the leading classical pipeline by 0.016 PR-AUC on both datasets. For two of the five individual classifiers on Bank Marketing the FT-Transformer does edge past that baseline. That margin is identical across two structurally very different problems and sits at the upper edge of the fold spread, which makes it the firmest quantitative statement the study supports.

The ordering within the classical family is a weaker result and is reported as such. Pipeline C obtained the highest PR-AUC on both datasets (0.4728 on Bank Marketing and 0.7797 on Online Retail), but its lead over the baseline is 0.0090 on Online Retail, where it holds for all five classifiers, against 0.0025 on Bank Marketing, where it holds for two of five. On Bank Marketing the two are not distinguishable and the comparison is left undecided.

Where the advantage of Pipeline C comes from is a separate question, and the answer is not the one its name suggests. The ablation attributes the entire gain to the polynomial interaction terms and none of it to the domain features, which reach 0.4121 against 0.4124 for raw input on Bank Marketing and reproduce the raw value exactly on Online Retail, where all seven pipelines share the same four RFM aggregates in any case. The result of this study is accordingly that interaction terms on standardized features pay off, not that domain knowledge does. Since constructing interaction terms requires no subject-matter expert, this

is the more widely applicable of the two readings.

A second observation limits the framing further. The spread within the classical family exceeded the spread between families, since the difference between the best and worst classical configuration was 0.144 PR-AUC on Bank Marketing, against 0.026 across all three AI pipelines. The relevant question is therefore not whether a pipeline is classical or learned, but which specific operations it performs.

The ablation study made this concrete. Standardization applied alone raised PR-AUC by 0.029 on Bank Marketing, which is three quarters of what the complete baseline pipeline achieved (0.040) and more than four of the seven complete pipelines gained over the raw input. Only two isolated operations surpassed it, interaction terms at 0.037 and the supervised encoder at 0.046. Several complete pipelines performed worse than their own best individual step, so the steps interact rather than accumulate. Pre-processing sophistication was not reliably rewarded.

Robustness was the one dimension on which the clean-data ordering did not carry over, and it is also where this work delivers less than it set out to, since a defect in the evaluation harness prevents the AI pipelines from being compared against the classical ones at all. The answer to RQ2 below is therefore given within each family rather than across them.

5.2 Research Questions Revisited

RQ1: Predictive performance. AI-based representation learning does not outperform classical pre-processing on either dataset. Pipeline C ranks first on both (0.4728 and 0.7797), the baseline second, and the best AI pipeline third, 0.016 PR-AUC behind the leader on both datasets. The first two places are separated by less than one fold standard deviation on Bank Marketing and their order is not a result there. The distance to the third is, and it is what answers the research question. The result is consistent across all five downstream classifiers, which indicates that it reflects a property of the representation rather than an interaction with a particular learner. It also connects to the established finding that tree-based classifiers retain an advantage over deep architectures on tabular data [7, 8, 45]. That comparison varies the classifier and holds the input fixed, whereas this one varies the input and holds the five classifiers fixed, so the same verdict is reached from the opposite direction, with the deep architecture acting as a feature extractor rather than as a classifier.

One objection has to be met here, since the downstream models were tuned while the encoder architectures were fixed a priori. The deficit of the AI pipelines could therefore in principle be an artifact of unequal tuning, but the ablation study speaks against that reading. It evaluates every representation with the same untuned logistic regression, giving no pipeline any tuning advantage at all, and the ordering it produces agrees with the tuned comparison: the encoders of Pipelines D and E gain 0.010 and lose 0.009 against raw input, while standardization alone gains 0.029. An unequal tuning budget cannot explain a deficit that persists when the budget is zero for everyone.

Among the learned representations, the supervised FT-Transformer outperformed both autoencoders on Bank Marketing and one of the two on Online Retail. Alignment between the encoder objective and the downstream task mattered more than architectural sophistication.

RQ2: Robustness. This question is answered on a narrower basis than intended. The robustness harness encodes with the single-pass encoders rather than the per-fold encoders of the corrected main experiment, so Pipelines D–F cannot be placed on a common scale with A–C, and no claim about the robustness of learned representations relative to classical pre-processing is made. The comparison within each family remains valid and is what follows.

No pipeline is robust in general. Among the classical pipelines the ordering of corruption mechanisms follows theory [22], with missingness not at random the most damaging, while completely random missingness and label noise are nearly harmless at the rates tested. Pipeline C is the systematic exception, since missingness at random costs it roughly twice as much: it removes values conditional on other observed columns and thereby strikes exactly the base features from which the derived features are computed. That same mechanism makes Pipeline C the least robust classical configuration on Bank Marketing, at 12.0 % mean degradation against 5.0 % for the baseline, so the property producing its clean-data advantage also produces its fragility. On Online Retail the effect disappears, where Pipeline C degrades by 3.0 % and ranks second behind the baseline, which locates the fragility in elaborate derived features on rich input rather than in the pipeline as such. Within the family of learned representations the training objective governs robustness far more than the architecture, the supervised FT-Transformer losing 6.6 % under MNAR against 27.7 % and 34.0 % for the two autoencoders.

Beyond these the result is dataset-dependent. PCA-based compression was the most stable classical configuration on Bank Marketing, discarding the variance directions in which injected noise lies, but the same mechanism was unavailable on Online Retail, where four input features leave no redundant directions and it became the least robust classical option instead.

A methodological caveat applies throughout, in that several degradation curves are non-monotonic, in some cases reaching their minimum at the lowest corruption rate tested. Robustness assessments based on a single corruption rate can therefore yield arbitrary orderings.

The practical implication is that the choice should be conditioned on data quality governance rather than on benchmark performance alone.

5.3 Practical Recommendation

Table 4.5 in Section 4.11 states the recommendation case by case. Its scope has to be carried along with it: two datasets do not establish generality, and of those two only Bank Marketing discriminates between the pipelines at all, since on Online Retail all seven fall within 0.017 PR-AUC of one another. What follows is therefore a guide for tabular binary classification on data of the size and structure examined here, not a general prescription. Three priorities condense it.

Correct execution of the elementary operations comes first, since imputation, encoding and above all scaling account for the majority of the achievable benefit and errors at this level do more damage than the absence of any sophisticated technique. Polynomial interaction terms on the standardized numeric features are the one elaboration that repaid its effort, and because that operation is generic it requires no subject-matter expert. It does require monitoring of the input features it is built on, which are the route by which corruption reaches it. Representation learning cannot be recommended for problems resembling those studied here, since it offers no accuracy advantage under a tuned downstream model and eliminates

the interpretability on which Business Intelligence decisions depend.

The last of these needs one qualification, because the ablation reverses the ordering: with the downstream model held fixed and untuned, the supervised encoder is the strongest single operation of the study. The advantage of the classical pipelines therefore rests on their responding better to downstream tuning, not on their carrying more information. Where a learned representation is nevertheless required, the supervised FT-Transformer is the only defensible option here, since the unsupervised autoencoders offered neither accuracy, nor interpretability, nor robustness.

Representation learning may become attractive at larger scale, with genuinely high dimensional or weakly structured input, or where the encoder can be pre-trained once and amortized across many downstream tasks. None of these conditions applied in this study.

5.4 Limitations

Two limitations concern the validity of the evaluation itself. The significance analysis applies a test that assumes independent observations to the dependent folds of a repeated cross-validation, so its p -values are not usable and no conclusion here rests on them. And the robustness harness encodes with single-pass rather than per-fold encoders, which invalidates every robustness comparison between the AI and the classical pipelines. Those claims have been withdrawn rather than repaired.

The remaining limitations concern scope. The findings rest on two datasets of moderate size, and the discriminating evidence comes essentially from Bank Marketing, since all seven pipelines fall within 0.017 PR-AUC of one another on Online Retail. The ablation employs logistic regression as a downstream proxy, and step contributions may differ for tree-based learners that are insensitive to feature scaling. Pipeline F sees the training labels during representation learning while the other six pipelines do not, an asymmetry of information that its defeat under the tuned protocol survives but that partly explains its victory in the ablation. Finally, compute budgets constrained the AI pipelines, since per-fold encoder training was capped at 60 epochs and hyper-parameter optimization at 20 trials per combination. A fuller treatment of all of these is given in Section 4.10.

5.5 Future Work

Three directions follow from these findings.

Broader empirical base. Extending the comparison to a standardized benchmark suite of tabular datasets would establish whether the observed ordering holds generally, and would allow the relationship between dataset characteristics and the value of pre-processing, hinted at here by the contrast between the feature-rich and the feature-sparse dataset, to be modeled explicitly.

Pre-training and transfer. The encoders in this study were trained from scratch on a single task, which is the least favorable setting for representation learning. Self-supervised pre-training on a larger corpus of related tabular data, with the representation subsequently transferred, would test the hypothesis that the observed deficit is a consequence of limited training signal rather than of the approach itself. The methods surveyed by Wang et al. [46]

and the synthetic-prior pre-training of Hollmann et al. [48] indicate that this route is now practicable.

Interpretability of learned representations. The interpretability gap identified here was measured only by the semantic emptiness of the latent axes. Applying attribution methods to map latent dimensions back onto original features would quantify how much of the gap is irreducible and how much is a tooling problem.

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List of Abbreviations

AI	Artificial Intelligence
BI	Business Intelligence
CPU	Central Processing Unit
CV	Cross-Validation
DAE	Denoising Autoencoder
FTT	Feature Tokenizer Transformer
HPO	Hyper-parameter Optimization
IQR	Interquartile Range
LGBM	LightGBM, a gradient boosting implementation
LR	Logistic Regression
MAR	Missing at Random
MCAR	Missing Completely at Random
MLP	Multi-Layer Perceptron
MNAR	Missing Not at Random
PCA	Principal Component Analysis
PR-AUC	Area Under the Precision–Recall Curve
RF	Random Forest
RFM	Recency, Frequency, Monetary value
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SMOTE	Synthetic Minority Over-sampling Technique
UCI	University of California, Irvine (Machine Learning Repository)
VAE	Variational Autoencoder
XGB	XGBoost, a gradient boosting implementation

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Appendix A

Appendix

A.1 Hyper-parameter Search Spaces

All downstream models were tuned with Optuna [39] using a tree-structured Parzen estimator with 20 trials per pipeline–model combination. The optimization objective was PR-AUC evaluated by an internal three-fold cross-validation on the training split. Table A.1 lists the search spaces.

Encoder configurations

The encoders of Pipelines D–F were not tuned by Optuna. Their configuration was fixed a priori in order to keep the compute budget tractable. Table A.2 documents the settings.

Table A.1: Hyper-parameter search spaces. “log” denotes logarithmic sampling.

Model	Parameter	Range	Scale
Logistic Regression	C	$10^{-4} - 100$	log
	l1_ratio	0.0 – 1.0	linear
	max_iter	500 – 5000	linear
Random Forest	n_estimators	100 – 1000	linear
	max_depth	3 – 30 or unlimited	linear
	min_samples_split	2 – 20	linear
	min_samples_leaf	1 – 10	linear
XGBoost	n_estimators	100 – 1000	linear
	max_depth	3 – 10	linear
	learning_rate	$10^{-4} - 0.3$	log
	subsample	0.5 – 1.0	linear
	colsample_bytree	0.5 – 1.0	linear
	reg_alpha	$10^{-8} - 10$	log
	reg_lambda	$10^{-8} - 10$	log
	min_child_weight	1 – 10	linear
LightGBM	num_leaves	20 – 300	linear
	max_depth	3 – 12	linear
	learning_rate	$10^{-4} - 0.3$	log
	n_estimators	100 – 1000	linear
	min_child_samples	5 – 100	linear
	subsample	0.5 – 1.0	linear
	colsample_bytree	0.5 – 1.0	linear
	reg_alpha	$10^{-8} - 10$	log
	reg_lambda	$10^{-8} - 10$	log
MLP	hidden_dim	32 – 512	linear
	n_layers	1 – 4	linear
	dropout	0.0 – 0.5	linear
	lr	$10^{-4} - 10^{-2}$	log
	batch_size	{64, 128, 256, 512}	categorical
	weight_decay	$10^{-6} - 10^{-2}$	log

Table A.2: Fixed encoder configurations for Pipelines D–F.

Pipeline	Parameter	Value
D: DAE	Encoder architecture	input $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 16
	Decoder architecture	mirrored
	Gaussian noise σ	0.2
	Masking probability	0.3
	Loss	MSE on clean input
	Latent dimension	16
E: VAE	Encoder architecture	input $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ $(\mu, \log \sigma^2)$
	KL weight β	1.0
	Representation	deterministic mean μ
	Latent dimension	16
F: FT-Transformer	Token dimension d	64
	Attention heads	8
	Encoder layers	3
	FFN factor	4/3
	Dropout	0.1
	Normalization	pre-layer norm
Common	Optimizer	Adam
	Batch size	256
	Max. epochs (single pass)	100
	Max. epochs (per CV fold)	60
	Early stopping	patience 10 on validation split (15 %)

Table A.3: Complete ablation on Bank Marketing. PR-AUC of every pre-processing step applied in isolation to the raw input, by pipeline, with logistic regression as the downstream model. Empty cells mean the step is not part of that pipeline. Raw baseline 0.4124.

Step	A	B	C	D	E	F
Raw input	0.4124	0.4124	0.4124	0.4124	0.4124	0.4124
Imputation	0.4124	0.4124	0.4124	0.4124	0.4124	0.4124
One-hot encoding	0.4098	0.4098	0.4098	0.4098	0.4098	0.4098
Winsorization	–	0.3884	–	–	–	–
Box-Cox	–	0.4394	–	–	–	–
Standardization	0.4415	0.4415	0.4415	0.4415	0.4415	0.4415
Domain features	–	–	0.4121	–	–	–
Interaction terms	–	–	0.4494	–	–	–
PCA 90 %	–	0.4044	–	–	–	–
PCA 20 %	–	0.3303	–	–	–	–
DAE	–	–	–	0.4223	–	–
VAE	–	–	–	–	0.4035	–
FT-Transformer	–	–	–	–	–	0.4586
Complete pipeline	0.4519	–	0.4508	0.4208	0.4064	0.4607
Complete, PCA 90 %	–	0.4185	–	–	–	–
Complete, PCA 20 %	–	0.2700	–	–	–	–

Table A.4: Complete ablation on Online Retail. PR-AUC of every pre-processing step applied in isolation to the raw input, by pipeline, with logistic regression as the downstream model. Empty cells mean the step is not part of that pipeline. Raw baseline 0.7350.

Step	A	B	C	D	E	F
Raw input	0.7350	0.7350	0.7350	0.7350	0.7350	0.7350
Imputation	0.7350	0.7350	0.7350	0.7350	0.7350	0.7350
One-hot encoding	0.7350	0.7350	0.7350	0.7350	0.7350	0.7350
Winsorization	–	0.7706	–	–	–	–
Box-Cox	–	0.7696	–	–	–	–
Standardization	0.7659	0.7659	0.7659	0.7659	0.7659	0.7659
Domain features	–	–	0.7350	–	–	–
Interaction terms	–	–	0.7666	–	–	–
PCA 90 %	–	0.7665	–	–	–	–
PCA 20 %	–	0.7554	–	–	–	–
DAE	–	–	–	0.7669	–	–
VAE	–	–	–	–	0.7635	–
FT-Transformer	–	–	–	–	–	0.7678
Complete pipeline	0.7658	–	0.7656	0.7669	0.7639	0.7661
Complete, PCA 90 %	–	0.7652	–	–	–	–
Complete, PCA 20 %	–	0.7636	–	–	–	–

A.2 Complete Ablation Results

Tables A.3 and A.4 give every value underlying Fig. 4.10 and Table 4.2. Each column is one pipeline and each row one operation applied on its own to the raw input, so that a column can be read as the decomposition of that pipeline.

Table A.5: Complete results, Bank Marketing. CV values are means over 15 folds.

Pipe	Model	PR-AUC	SD	ROC-AUC	F_1	Brier	Lift	Test
A	LR	0.4497	0.0145	0.7897	0.3365	0.0789	4.32	0.4652
	RF	0.4678	0.0138	0.7967	0.3467	0.0772	4.52	0.4936
	XGB	0.4679	0.0146	0.7991	0.3493	0.0772	4.54	0.4935
	LGBM	0.4703	0.0150	0.8002	0.3579	0.0770	4.54	0.4933
	MLP	0.4492	0.0136	0.7891	0.4544	0.1521	4.34	0.4724
B ₉₀	LR	0.4135	0.0145	0.7804	0.3028	0.0823	4.11	0.4421
	RF	0.4224	0.0128	0.7811	0.2967	0.0811	4.09	0.4533
	XGB	0.4264	0.0121	0.7815	0.3062	0.0808	4.14	0.4573
	LGBM	0.4257	0.0126	0.7810	0.3018	0.0808	4.16	0.4497
	MLP	0.4221	0.0199	0.7830	0.4434	0.1671	4.13	0.4547
B ₂₀	LR	0.2683	0.0090	0.7366	0.0000	0.0916	2.76	0.2784
	RF	0.3285	0.0104	0.7588	0.1356	0.0874	3.58	0.3659
	XGB	0.3244	0.0121	0.7583	0.0498	0.0878	3.49	0.3486
	LGBM	0.3253	0.0108	0.7578	0.1119	0.0877	3.50	0.3474
	MLP	0.3165	0.0109	0.7554	0.3538	0.1984	3.37	0.3233
C	LR	0.4496	0.0142	0.7901	0.3320	0.0789	4.33	0.4620
	RF	0.4698	0.0136	0.8006	0.3608	0.0770	4.54	0.4939
	XGB	0.4728	0.0129	0.8020	0.3292	0.0769	4.55	0.4923
	LGBM	0.4696	0.0133	0.7998	0.3573	0.0771	4.53	0.4946
	MLP	0.4458	0.0129	0.7856	0.4451	0.1700	4.36	0.4777
D	LR	0.4185	0.0155	0.7735	0.3104	0.0813	4.17	0.4549
	RF	0.4255	0.0122	0.7768	0.2891	0.0808	4.19	0.4429
	XGB	0.4309	0.0145	0.7796	0.3140	0.0803	4.24	0.4573
	LGBM	0.4305	0.0145	0.7791	0.2994	0.0804	4.21	0.4539
	MLP	0.4281	0.0154	0.7847	0.4386	0.1732	4.17	0.4695
E	LR	0.4082	0.0196	0.7713	0.2798	0.0821	4.14	0.4370
	RF	0.4342	0.0141	0.7840	0.3108	0.0799	4.21	0.4622
	XGB	0.4329	0.0136	0.7849	0.0310	0.0840	4.23	0.4664
	LGBM	0.4302	0.0129	0.7838	0.2580	0.0807	4.18	0.4586
	MLP	0.4221	0.0159	0.7782	0.4143	0.1697	4.18	0.4445
F	LR	0.4541	0.0202	0.7901	0.3693	0.0784	4.42	0.4750
	RF	0.4558	0.0189	0.7896	0.3412	0.0781	4.43	0.4716
	XGB	0.4572	0.0173	0.7908	0.3302	0.0783	4.41	0.4780
	LGBM	0.4554	0.0185	0.7896	0.2717	0.0791	4.42	0.4779
	MLP	0.4543	0.0190	0.7896	0.4582	0.1623	4.41	0.4781

A.3 Complete Results

Tables A.5 and A.6 report all metrics for every pipeline–model combination. Cross-validation values are means over 15 folds, while the final column is the held-out test split, provided as an independent check.

Table A.6: Complete results, Online Retail. CV values are means over 15 folds.

Pipe	Model	PR-AUC	SD	ROC-AUC	F_1	Brier	Lift	Test
A	LR	0.7648	0.0180	0.7330	0.6560	0.2076	1.85	0.7747
	RF	0.7721	0.0193	0.7378	0.6525	0.2046	1.86	0.7739
	XGB	0.7698	0.0193	0.7379	0.6586	0.2209	1.85	0.7692
	LGBM	0.7664	0.0170	0.7326	0.6585	0.2083	1.87	0.7611
	MLP	0.7677	0.0214	0.7334	0.6336	0.2073	1.85	0.7780
B ₉₀	LR	0.7637	0.0173	0.7310	0.6747	0.2102	1.81	0.7707
	RF	0.7583	0.0184	0.7264	0.6291	0.2101	1.82	0.7706
	XGB	0.7470	0.0187	0.7114	0.6458	0.2135	1.81	0.7612
	LGBM	0.7501	0.0202	0.7206	0.6468	0.2108	1.79	0.7629
	MLP	0.7615	0.0167	0.7312	0.6738	0.2124	1.81	0.7687
B ₂₀	LR	0.7627	0.0174	0.7292	0.6730	0.2105	1.82	0.7686
	RF	0.7558	0.0178	0.7210	0.6382	0.2100	1.82	0.7643
	XGB	0.7564	0.0187	0.7208	0.6483	0.2189	1.82	0.7639
	LGBM	0.7472	0.0184	0.7142	0.6440	0.2122	1.81	0.7541
	MLP	0.7622	0.0178	0.7279	0.6349	0.2099	1.82	0.7653
C	LR	0.7766	0.0169	0.7431	0.6611	0.2023	1.83	0.7922
	RF	0.7789	0.0193	0.7466	0.6611	0.2018	1.87	0.7840
	XGB	0.7797	0.0179	0.7503	0.6641	0.2028	1.86	0.7845
	LGBM	0.7757	0.0196	0.7435	0.6712	0.2188	1.86	0.7805
	MLP	0.7750	0.0166	0.7420	0.6443	0.2052	1.85	0.7804
D	LR	0.7629	0.0194	0.7303	0.6432	0.2071	1.85	0.7748
	RF	0.7641	0.0178	0.7341	0.6517	0.2064	1.84	0.7748
	XGB	0.7631	0.0183	0.7325	0.6502	0.2070	1.85	0.7716
	LGBM	0.7601	0.0194	0.7315	0.6651	0.2263	1.83	0.7707
	MLP	0.7628	0.0208	0.7314	0.6414	0.2141	1.84	0.7737
E	LR	0.7595	0.0222	0.7280	0.6535	0.2089	1.83	0.7709
	RF	0.7612	0.0187	0.7314	0.6444	0.2071	1.83	0.7674
	XGB	0.7613	0.0193	0.7319	0.6532	0.2156	1.83	0.7665
	LGBM	0.7593	0.0194	0.7304	0.6625	0.2327	1.84	0.7654
	MLP	0.7623	0.0219	0.7300	0.6519	0.2117	1.84	0.7681
F	LR	0.7633	0.0191	0.7301	0.6389	0.2072	1.85	0.7764
	RF	0.7634	0.0170	0.7323	0.6497	0.2062	1.85	0.7723
	XGB	0.7635	0.0172	0.7314	0.6495	0.2072	1.86	0.7668
	LGBM	0.7626	0.0186	0.7320	0.6590	0.2172	1.85	0.7686
	MLP	0.7607	0.0217	0.7282	0.6197	0.2097	1.83	0.7771

Table A.7: PR-AUC under MNAR missingness at increasing corruption rates. Several curves reach their minimum at low rates and partially recover, which is why Table 4.3 reports means over all rates.

Pipeline	Baseline	5 %	10 %	20 %	30 %	50 %
<i>Bank Marketing</i>						
A	0.452	0.437	0.430	0.407	0.405	0.395
B ₉₀	0.419	0.424	0.424	0.423	0.423	0.389
B ₂₀	0.270	0.322	0.288	0.346	0.347	0.245
C	0.451	0.344	0.379	0.379	0.377	0.395
D	0.427	0.293	0.300	0.327	0.326	0.300
E	0.405	0.223	0.244	0.316	0.321	0.233
F	0.476	0.458	0.455	0.458	0.456	0.395
<i>Online Retail</i>						
A	0.766	0.763	0.762	0.743	0.668	0.712
B ₉₀	0.765	0.766	0.763	0.707	0.645	0.570
B ₂₀	0.764	0.761	0.759	0.739	0.722	0.570
C	0.766	0.765	0.758	0.650	0.623	0.690
D	0.766	0.758	0.756	0.741	0.574	0.549
E	0.766	0.759	0.761	0.756	0.700	0.552
F	0.770	0.765	0.761	0.701	0.734	0.536

A.4 Robustness: Full Degradation Curves

Table A.7 shows the complete MNAR degradation curves, which document the non-monotonic behavior discussed in Section 4.5.

A.5 Pipeline and Corruption Pseudocode

Chapter 3 describes the pipelines at the level of their constituent operations. This section states the operations that carry a design decision precisely enough to be re-implemented without consulting the repository. Pipelines A and B are omitted, since they are compositions of standard scikit-learn transformers with the parameters given in Section 3.3.

Pipeline C: derived features and interaction terms

Four domain features are defined for Bank Marketing:

```
age_x_campaign          = age * campaign
contact_intensity       = campaign / (pdays + 1)
was_previously_contacted = 1 if pdays != 999 else 0
previous_x_poutcome_success = previous * (poutcome == "success")
```

The addition of 1 in the denominator of `contact_intensity` avoids division by zero at `pdays = 0`. The flag `was_previously_contacted` decodes the sentinel 999, which Section 4.10 discusses as the one domain quirk that only Pipeline C resolves.

On Online Retail no separate domain features are constructed. The RFM aggregation is performed by the shared loader and is therefore present in every pipeline, which is the reason the ablation records a contribution of exactly zero for the domain step on that dataset.

The interaction terms are generated by `PolynomialFeatures` with degree 2 and `interaction_`

only=True, applied after imputation and before scaling, and fitted on the training fold only.

```
num_cols = all numeric columns except the target
for each unordered pair (i, j) with i < j in num_cols:
    new column x_i * x_j
categorical columns pass through unchanged
no bias term is added; centring is left to the scaler
```

Neither the participating features nor the pairs are selected: every pair of numeric columns is formed. On Bank Marketing this expands 62 columns to 144.

Pipelines D, E and F: encoder input and fitting

All three encoders receive numeric features standardized and categorical features label encoded, except that Pipeline F embeds categoricals through a learned lookup table instead. The architectures and their fixed hyper-parameters are listed in Table A.2.

The fitting protocol is what distinguishes the corrected main experiment from the standalone pipeline runs, and it is the point at which the leakage described in Section 3.6.3 arises or is avoided.

```
for each of the 15 cross-validation splits (train_idx, val_idx):
    fit encoder on X[train_idx] only
    early stopping on a held-out part of train_idx,
    not on the training loss
    Z_train = encode(X[train_idx])
    Z_val = encode(X[val_idx])
    cache (Z_train, Z_val) so that all five downstream
    models reuse the identical encoding
```

Fitting the encoder once on the whole training split and cross-validating on its output, which is what the standalone scripts do, means every validation fold has already been seen by the encoder, and for the supervised Pipeline F including its labels.

Corruption mechanisms

Corruption is applied to the training fold only; the validation fold stays clean. All mechanisms operate on numeric columns. A column whose missing share exceeds 50 % is re-inserted as missing rather than dropped, since a deployed encoder cannot change its input dimension at inference time.

```
MCAR(X, rate):
    mask a uniformly random 'rate' fraction of all numeric cells

MAR(X, rate):
    for each numeric column c:
        pivot <- a randomly chosen numeric column != c
        t <- quantile(X[pivot], 1 - rate)
        set X[c] to missing wherever X[pivot] >= t
    (falls back to MCAR when fewer than two numeric columns exist)

MNAR(X, rate):
    for each numeric column c:
        t <- quantile(X[c], 1 - rate)
        set X[c] to missing wherever X[c] >= t
```

```

OUTLIER(X, rate, k = 5):
  for each numeric column c:
    q1, q3 <- 25th and 75th percentile of c
    iqr <- q3 - q1 (fallback to sd(c) if iqr == 0)
    draw ceil(rate * n) row indices without replacement
    replace each with q3 + k*iqr + U(0, iqr) with probability 1/2
                      or q1 - k*iqr - U(0, iqr) otherwise

LABEL_NOISE(y, rate):
  flip each label independently with probability 'rate'
  (symmetric mode; both classes are affected)

```

The MAR mechanism removes values conditional on another observed column, which is why it strikes the base features of Pipeline C so much harder than MNAR does. Missing values are subsequently imputed by the median of the training fold, which is unbiased under MCAR, approximately correctable under MAR, and systematically biased under MNAR.

Temporal split of Online Retail

-

```

cutoff = 2011-10-01
feature period : all invoices with InvoiceDate < cutoff
target period  : all invoices with InvoiceDate >= cutoff

per customer, computed on the feature period only:
  Recency      = (cutoff - date of last purchase) in days
  Frequency    = number of distinct invoices
  Monetary     = total revenue
  TotalItems   = total quantity

target = 1 if the customer transacts at least once
        in the target period, else 0

```

The cutoff places roughly the final quarter of the observation period into the target window, which leaves ten months of history for the aggregates while keeping the target window long enough for the positive rate to reach 50.8 %. Because features and target are separated in time, the target cannot be a deterministic function of its own inputs, which is the leakage that a “high-value customer” label derived from **Monetary** would have introduced.

A.6 Reproducibility and Code Availability

The complete implementation is published in a public repository. Table A.8 summarizes its structure.

A single random seed (42) governs NumPy, PyTorch and all scikit-learn splits [51]. A complete run (70 experiments plus ablation, robustness sweep, learning curves and all figures) requires approximately eight to eleven hours on CPU at the hyper-parameter budget used.

The datasets are not redistributed. They are obtained from the UCI Machine Learning Repository [49, 50] and placed in `data/raw/`.

Table A.8: Repository structure.

Path	Content
<code>config.py</code>	Seed, fold counts, HPO budget, paths
<code>scripts/eda/</code>	Exploratory analysis
<code>scripts/classical/</code>	Pipelines A–C and shared transformers
<code>scripts/ai/</code>	Pipelines D–F (PyTorch)
<code>scripts/eval/</code>	CV runner, metrics, model factories, result tables
<code>scripts/robustness/</code>	Corruption injection and robustness sweep
<code>scripts/visualize/</code>	Figure generation
<code>scripts/ablation.py</code>	Ablation study
<code>scripts/run_experiment.py</code>	Main experiment orchestration

Repository: <https://github.com/poppdan/Systematic-Comparison-Pre-processing-for-Business-Intelligence>